MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
49PS POWER GENERATION:
SOLAR PLANT
DESIGN OF FOUNDATION
FOR MODULE STRUCTURE

Document Number: **MD-216-49PS-EG-CV-CAL-1604**
Revision: **D01**
Revision Date: **30.10.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	WG02	CED01	45005	OU100688098	1
L&T Document Number:	O19045	008	C	K	24300	001	1

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	29.08.2020	FIRST ISSUE	SG	AVG	SJM	
C01	29.09.2020	ISSUED FOR CONSTRUCTION	SG	AVG	SJM	
D01	30.10.2023	AS BUILT	SG	AVG	SJM	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1. INTENT 4
2. EXECUTIVE SUMMARY 4
3. SCOPE 4
4. MODULE MOUNTING STRUCTURE DESIGN DIAGRAMS 5
   - 4.1 Model Geometry 5
   - 4.2 Model MMS Post Supports 6
5. MMS SUPPORT REACTIONS 6
   - 5.1 Model Supports 6
   - 5.2 Basic Load Cases 7
   - 5.3 Combination of Load Cases 7
   - 5.3 Load Cases Reaction Forces Summary 8
   - 5.4 Load Combinations Reaction Forces Summary 10
   - 5.5 Foundation Design Forces 23
6. DESIGN OF MMS CONCRETE PILE FOUNDATION 24
7. DESIGN OF MMS FOUNDATION WITH STAAD PRO 27
   - 7.1. Soil Spring Calculation 27
   - 7.2. Pile Deflection Graph 28
   - 7.3. Calculation of Effective Compression Height at Base 29
8. SOIL DESIGN PARAMETERS FROM GEOTECH REPORT 30

References 31

ATTACHMENT 1 STAAD INPUT

ATTACHMENT 2 FORMULA REFERENCE PAGES

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

The document is prepared for the design calculation of the foundation system for the solar module mounting structure. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughout capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

Chapter 14 & 16 of the SBC 306 deals with piled foundations from the aspects of general requirements for pile foundations, material requirements actions (forces) on piles from superimposed loading or ground movements, structural design and supervision of construction, SBC 306 Code does not make recommendations on methods of pile design.

The document is prepared based on the BS 8004- "Code of practice for Foundations" and "Pile Design and Construction Practice By M.J.Tomlinson" the book is more towards the international eurocode. The foundation design also checked using the STAAD software based on the available soil report properties.

Following documents are referred for calculating the foundation depth.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

a. MD-216-49PS-EG-CV-GAD-1603 MOUDULE MOUNTING STRUCTURE FOUNDATION LAYOUT

b. MD-216-49PS-EG-CV-GAD-1604 MODULE MOUNTING STRUCTURE FOUDATION CROSS SECTION AND DETAILS

c. MD-216-49PS-EG-ST-CAL-2415 DESIGN CALCULATIONS OF MODULE MOUNTING STRUCTRE

4. MODULE MOUNTING STRUCTURE DESIGN DIAGRAMS

4.1 Model Geometry

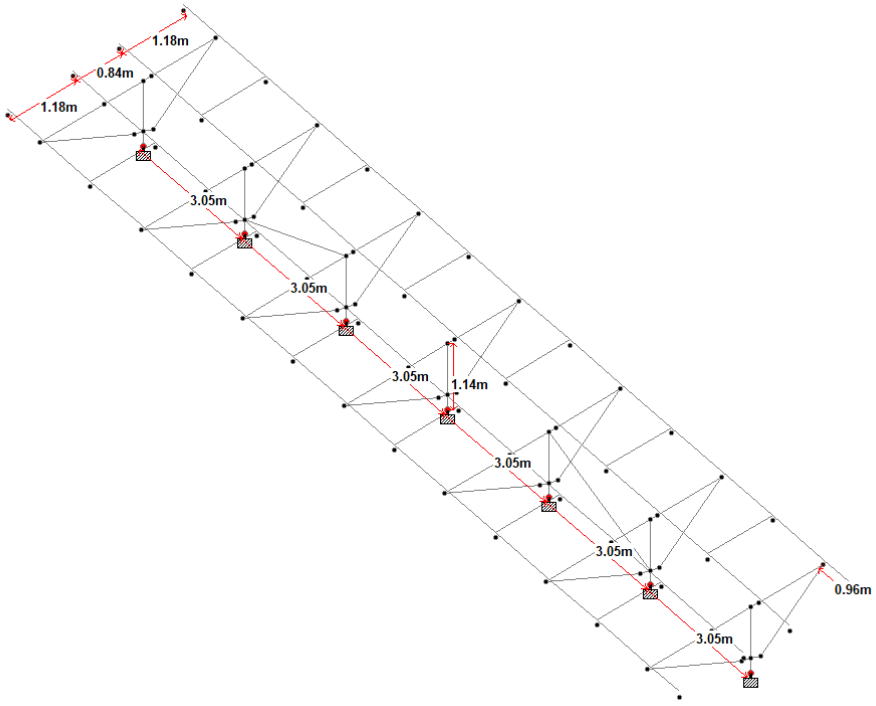


Figure 1: STAAD Model Geometry

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.2 Model MMS Post Supports

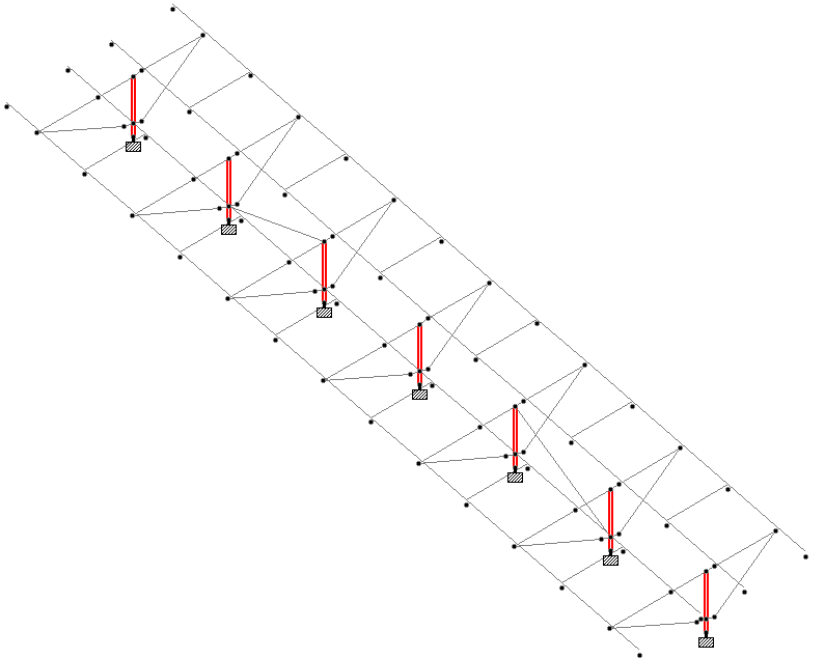


Figure 2: STAAD Model MMS Post Supports

5. MMS SUPPORT REACTIONS

5.1 Model Supports

Node	X (kN/mm)	Y (kN/mm)	Z (kN/mm)	rX (kN-m/deg)	rY (kN-m/deg)	rZ (kN-m/deg)
51	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
52	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
53	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

54	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
55	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
56	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
57	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed

5.2 Basic Load Cases

Load Case	Load Detail
1	DEAD LOAD
2	WIND LOAD UPLIFT
3	WIND LOAD PUSH
4	WIND LOAD +X
5	WIND LOAD +Z
6	WIND LOAD -X
7	WIND LOAD -Z

5.3 Combination of Load Cases

As per SBC – 301(2.4.1 Basic Combinations).

Type	LC	Name
Combination	101	D.L.
Combination	102	D.L.+ W.L(UPLIFT)+ W.L.(+X)
Combination	103	D.L.+ W.L(UPLIFT)+ W.L.(-X)
Combination	104	D.L.+ W.L(PUSH)+ W.L.(+X)
Combination	105	D.L.+ W.L(PUSH)+ W.L.(-X)
Combination	106	D.L.+ W.L(UPLIFT)+ W.L.(+Z)
Combination	107	D.L.+ W.L(UPLIFT)+ W.L.(-Z)
Combination	108	D.L.+ W.L(PUSH)+ W.L.(+Z)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Combination	109	D.L.+ W.L(PUSH)+ W.L.(-Z)
Combination	110	D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)
Combination	111	D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)
Combination	112	D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)
Combination	113	D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)
Combination	114	D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)
Combination	115	D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)
Combination	116	D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)
Combination	117	D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)
Combination	118	0.6 D.L.+ W.L(UPLIFT)+ W.L.(+X)
Combination	119	0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)
Combination	120	0.6 D.L.+ W.L(PUSH)+ W.L.(+X)
Combination	121	0.6 D.L.+ W.L(PUSH)+ W.L.(-X)
Combination	122	0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)
Combination	123	0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)
Combination	124	0.6 D.L.+ W.L(PUSH)+ W.L.(+X)
Combination	125	0.6 D.L.+ W.L(PUSH)+ W.L.(-X)

5.3 Load Cases Reaction Forces Summary

Node	L/C	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
51	1 DEAD LOAD	0.02	1.61	0.01	0.01	0.00	-0.45
51	2 WIND LOAD UPLIFT	2.28	-7.45	-0.03	-0.03	0.00	-0.52
51	3 WIND LOAD PUSH	-1.43	4.67	0.02	0.02	0.00	0.33
51	4 WIND LOAD +X	0.53	0.00	0.00	0.00	0.00	-0.33
51	5 WIND LOAD +Z	-0.03	-0.02	0.29	0.09	0.00	0.03
51	6 WIND LOAD -X	-0.53	0.00	0.00	0.00	0.00	0.33
51	7 WIND LOAD -Z	0.03	0.02	-0.29	-0.09	0.00	-0.03
52	1 DEAD LOAD	-0.03	2.15	0.02	0.00	0.00	-0.53
52	2 WIND LOAD UPLIFT	3.10	-10.18	-0.08	-0.01	0.00	-0.73

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

52	3 WIND LOAD PUSH	-1.95	6.38	0.05	0.01	0.00	0.46
52	4 WIND LOAD +X	0.71	0.00	0.00	0.00	0.00	-0.40
52	5 WIND LOAD +Z	0.00	0.32	1.42	0.30	0.00	0.00
52	6 WIND LOAD -X	-0.71	0.00	0.00	0.00	0.00	0.40
52	7 WIND LOAD -Z	0.00	-0.32	-1.42	-0.30	0.00	0.00
53	1 DEAD LOAD	0.01	2.03	0.00	0.00	0.00	-0.55
53	2 WIND LOAD UPLIFT	2.91	-9.48	0.00	0.00	0.00	-0.67
53	3 WIND LOAD PUSH	-1.82	5.94	0.00	0.00	0.00	0.42
53	4 WIND LOAD +X	0.69	0.00	0.00	0.00	0.00	-0.49
53	5 WIND LOAD +Z	0.00	-0.30	0.28	0.08	0.00	0.00
53	6 WIND LOAD -X	-0.69	0.00	0.00	0.00	0.00	0.49
53	7 WIND LOAD -Z	0.00	0.30	-0.28	-0.08	0.00	0.00
54	1 DEAD LOAD	0.00	2.04	0.00	0.00	0.00	-0.54
54	2 WIND LOAD UPLIFT	2.97	-9.70	0.00	0.00	0.00	-0.69
54	3 WIND LOAD PUSH	-1.86	6.08	0.00	0.00	0.00	0.43
54	4 WIND LOAD +X	0.54	0.00	0.00	0.00	0.00	-0.38
54	5 WIND LOAD +Z	0.00	0.00	0.29	0.09	0.00	0.00
54	6 WIND LOAD -X	-0.54	0.00	0.00	0.00	0.00	0.38
54	7 WIND LOAD -Z	0.00	0.00	-0.29	-0.09	0.00	0.00
55	1 DEAD LOAD	0.01	2.03	0.00	0.00	0.00	-0.55
55	2 WIND LOAD UPLIFT	2.91	-9.48	0.00	0.00	0.00	-0.67
55	3 WIND LOAD PUSH	-1.82	5.94	0.00	0.00	0.00	0.42
55	4 WIND LOAD +X	0.69	0.00	0.00	0.00	0.00	-0.49
55	5 WIND LOAD +Z	0.00	0.30	0.28	0.08	0.00	0.00
55	6 WIND LOAD -X	-0.69	0.00	0.00	0.00	0.00	0.49
55	7 WIND LOAD -Z	0.00	-0.30	-0.28	-0.08	0.00	0.00
56	1 DEAD LOAD	-0.03	2.15	-0.02	0.00	0.00	-0.53
56	2 WIND LOAD UPLIFT	3.10	-10.18	0.08	0.01	0.00	-0.73

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

56	3 WIND LOAD PUSH	-1.95	6.38	-0.05	-0.01	0.00	0.46
56	4 WIND LOAD +X	0.71	0.00	0.00	0.00	0.00	-0.40
56	5 WIND LOAD +Z	0.00	-0.32	1.42	0.30	0.00	0.00
56	6 WIND LOAD -X	-0.71	0.00	0.00	0.00	0.00	0.40
56	7 WIND LOAD -Z	0.00	0.32	-1.42	-0.30	0.00	0.00
57	1 DEAD LOAD	0.02	1.61	-0.01	-0.01	0.00	-0.45
57	2 WIND LOAD UPLIFT	2.28	-7.45	0.03	0.03	0.00	-0.52
57	3 WIND LOAD PUSH	-1.43	4.67	-0.02	-0.02	0.00	0.33
57	4 WIND LOAD +X	0.53	0.00	0.00	0.00	0.00	-0.33
57	5 WIND LOAD +Z	0.03	0.02	0.29	0.09	0.00	-0.03
57	6 WIND LOAD -X	-0.53	0.00	0.00	0.00	0.00	0.33
57	7 WIND LOAD -Z	-0.03	-0.02	-0.29	-0.09	0.00	0.03
51	1 DEAD LOAD	0.02	1.61	0.01	0.01	0.00	-0.45
51	2 WIND LOAD UPLIFT	2.28	-7.45	-0.03	-0.03	0.00	-0.52
51	3 WIND LOAD PUSH	-1.43	4.67	0.02	0.02	0.00	0.33
51	4 WIND LOAD +X	0.53	0.00	0.00	0.00	0.00	-0.33
51	5 WIND LOAD +Z	-0.03	-0.02	0.29	0.09	0.00	0.03
51	6 WIND LOAD -X	-0.53	0.00	0.00	0.00	0.00	0.33
51	7 WIND LOAD -Z	0.03	0.02	-0.29	-0.09	0.00	-0.03

5.4 Load Combinations Reaction Forces Summary

Node	L/C	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
51	101 D.L.	0.03	2.26	0.01	0.01	0.00	-0.63
51	102 D.L.+ W.L(UPLIFT)+ W.L.(+X)	2.83	-5.83	-0.02	-0.03	0.00	-1.31

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

51	103 D.L.+ W.L(UPLIFT)+ W.L.(-X)	1.77	-5.83	-0.02	-0.03	0.00	-0.65
51	104 D.L.+ W.L(PUSH)+ W.L.(+X)	-0.88	6.28	0.02	0.03	0.00	-0.45
51	105 D.L.+ W.L(PUSH)+ W.L.(-X)	-1.93	6.28	0.02	0.03	0.00	0.21
51	106 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.27	-5.86	0.27	0.07	0.00	-0.95
51	107 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.32	-5.81	-0.32	-0.12	0.00	-1.01
51	108 D.L.+ W.L(PUSH)+ W.L.(+Z)	-1.43	6.26	0.32	0.12	0.00	-0.09
51	109 D.L.+ W.L(PUSH)+ W.L.(-Z)	-1.38	6.31	-0.27	-0.07	0.00	-0.15
51	110 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)	2.13	-3.97	-0.02	-0.02	0.00	-1.09
51	111 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)	1.33	-3.97	-0.02	-0.02	0.00	-0.60
51	112 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.65	5.12	0.02	0.02	0.00	-0.45
51	113 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.45	5.12	0.02	0.02	0.00	0.04
51	114 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)	1.71	-3.99	0.21	0.05	0.00	-0.82
51	115 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)	1.75	-3.95	-0.24	-0.09	0.00	-0.87

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

51	116 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.65	5.12	0.02	0.02	0.00	-0.45
51	117 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.45	5.12	0.02	0.02	0.00	0.04
51	118 0.6 D.L.+ W.L(UPLIFT)+ 1W.L.(+X)	2.82	-6.48	-0.02	-0.03	0.00	-1.13
51	119 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)	1.76	-6.48	-0.02	-0.03	0.00	-0.47
51	120 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-0.89	5.64	0.02	0.02	0.00	-0.27
51	121 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-1.94	5.64	0.02	0.02	0.00	0.39
51	122 0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.26	-6.50	0.27	0.06	0.00	-0.77
51	123 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.31	-6.46	-0.32	-0.12	0.00	-0.83
51	124 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-0.89	5.64	0.02	0.02	0.00	-0.27
51	125 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-1.94	5.64	0.02	0.02	0.00	0.39
52	101 D.L.	-0.04	3.01	0.02	0.00	0.00	-0.74
52	102 D.L.+ W.L(UPLIFT)+ W.L.(+X)	3.78	-8.02	-0.06	-0.01	0.00	-1.66
52	103 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.37	-8.03	-0.07	-0.01	0.00	-0.86
52	104 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.27	8.53	0.07	0.01	0.00	-0.47

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

52	105 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.68	8.53	0.07	0.01	0.00	0.32
52	106 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	3.07	-7.70	1.35	0.30	0.00	-1.26
52	107 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	3.08	-8.35	-1.48	-0.31	0.00	-1.26
52	108 D.L.+ W.L(PUSH)+ W.L.(+Z)	-1.98	8.86	1.49	0.31	0.00	-0.07
52	109 D.L.+ W.L(PUSH)+ W.L.(-Z)	-1.97	8.21	-1.35	-0.29	0.00	-0.08
52	110 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)	2.83	-5.48	-0.04	0.00	0.00	-1.37
52	111 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)	1.77	-5.48	-0.05	-0.01	0.00	-0.78
52	112 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.96	6.94	0.06	0.01	0.00	-0.49
52	113 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-2.02	6.94	0.05	0.01	0.00	0.11
52	114 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)	2.30	-5.24	1.02	0.22	0.00	-1.07
52	115 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)	2.30	-5.72	-1.11	-0.23	0.00	-1.08
52	116 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.96	6.94	0.06	0.01	0.00	-0.49
52	117 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-2.02	6.94	0.05	0.01	0.00	0.11

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

52	118 0.6 D.L.+ W.L(UPLIFT)+ 1W.L.(+X)	3.79	-8.88	-0.07	-0.01	0.00	-1.44
52	119 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.38	-8.89	-0.07	-0.01	0.00	-0.65
52	120 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.26	7.67	0.06	0.01	0.00	-0.26
52	121 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.67	7.67	0.06	0.01	0.00	0.54
52	122 0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	3.08	-8.56	1.35	0.29	0.00	-1.04
52	123 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	3.09	-9.21	-1.49	-0.31	0.00	-1.05
52	124 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.26	7.67	0.06	0.01	0.00	-0.26
52	125 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.67	7.67	0.06	0.01	0.00	0.54
53	101 D.L.	0.01	2.84	0.00	0.00	0.00	-0.77
53	102 D.L.+ W.L(UPLIFT)+ W.L.(+X)	3.61	-7.45	0.00	0.00	0.00	-1.71
53	103 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.22	-7.45	0.00	0.00	0.00	-0.73
53	104 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.12	7.96	0.00	0.00	0.00	-0.62
53	105 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.51	7.97	0.00	0.00	0.00	0.36
53	106 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.91	-7.76	0.28	0.08	0.00	-1.22

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

53	107 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.91	-7.15	-0.28	-0.08	0.00	-1.22
53	108 D.L.+ W.L(PUSH)+ W.L.(+Z)	-1.81	7.66	0.28	0.08	0.00	-0.13
53	109 D.L.+ W.L(PUSH)+ W.L.(-Z)	-1.82	8.27	-0.28	-0.08	0.00	-0.13
53	110 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)	2.71	-5.08	0.00	0.00	0.00	-1.42
53	111 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)	1.67	-5.08	0.00	0.00	0.00	-0.68
53	112 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.84	6.48	0.00	0.00	0.00	-0.60
53	113 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.88	6.48	0.00	0.00	0.00	0.14
53	114 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)	2.19	-5.31	0.21	0.06	0.00	-1.05
53	115 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)	2.19	-4.85	-0.21	-0.06	0.00	-1.05
53	116 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.84	6.48	0.00	0.00	0.00	-0.60
53	117 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.88	6.48	0.00	0.00	0.00	0.14
53	118 0.6 D.L.+ W.L(UPLIFT)+ 1W.L.(+X)	3.60	-8.26	0.00	0.00	0.00	-1.49
53	119 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.22	-8.26	0.00	0.00	0.00	-0.51

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

53	120 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.12	7.15	0.00	0.00	0.00	-0.40
53	121 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.51	7.16	0.00	0.00	0.00	0.58
53	122 0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.91	-8.57	0.28	0.08	0.00	-1.00
53	123 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.91	-7.96	-0.28	-0.08	0.00	-1.00
53	124 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.12	7.15	0.00	0.00	0.00	-0.40
53	125 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.51	7.16	0.00	0.00	0.00	0.58
54	101 D.L.	0.00	2.86	0.00	0.00	0.00	-0.76
54	102 D.L.+ W.L(UPLIFT)+ W.L.(+X)	3.50	-7.65	0.00	0.00	0.00	-1.61
54	103 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.43	-7.66	0.00	0.00	0.00	-0.85
54	104 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.33	8.13	0.00	0.00	0.00	-0.49
54	105 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.40	8.12	0.00	0.00	0.00	0.27
54	106 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.96	-7.65	0.29	0.09	0.00	-1.23
54	107 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.96	-7.65	-0.29	-0.09	0.00	-1.23
54	108 D.L.+ W.L(PUSH)+ W.L.(+Z)	-1.86	8.12	0.29	0.09	0.00	-0.11

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

54	109 D.L.+ W.L(PUSH)+ W.L.(-Z)	-1.86	8.12	-0.29	-0.09	0.00	-0.11
54	110 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)	2.62	-5.23	0.00	0.00	0.00	-1.34
54	111 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)	1.82	-5.23	0.00	0.00	0.00	-0.77
54	112 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-1.00	6.61	0.00	0.00	0.00	-0.50
54	113 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.80	6.60	0.00	0.00	0.00	0.07
54	114 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)	2.22	-5.23	0.22	0.07	0.00	-1.06
54	115 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)	2.22	-5.23	-0.22	-0.07	0.00	-1.06
54	116 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-1.00	6.61	0.00	0.00	0.00	-0.50
54	117 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.80	6.60	0.00	0.00	0.00	0.07
54	118 0.6 D.L.+ W.L(UPLIFT)+ 1W.L.(+X)	3.50	-8.47	0.00	0.00	0.00	-1.39
54	119 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.43	-8.47	0.00	0.00	0.00	-0.63
54	120 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.32	7.31	0.00	0.00	0.00	-0.27
54	121 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.40	7.30	0.00	0.00	0.00	0.49

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

54	122 0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.96	-8.47	0.29	0.09	0.00	-1.01
54	123 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.96	-8.47	-0.29	-0.09	0.00	-1.01
54	124 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.32	7.31	0.00	0.00	0.00	-0.27
54	125 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.40	7.30	0.00	0.00	0.00	0.49
55	101 D.L.	0.01	2.84	0.00	0.00	0.00	-0.77
55	102 D.L.+ W.L(UPLIFT)+ W.L.(+X)	3.61	-7.45	0.00	0.00	0.00	**-1.71**
55	103 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.22	-7.45	0.00	0.00	0.00	-0.73
55	104 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.12	7.96	0.00	0.00	0.00	-0.62
55	105 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.51	7.97	0.00	0.00	0.00	0.36
55	106 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.91	-7.15	0.28	0.08	0.00	-1.22
55	107 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.91	-7.76	-0.28	-0.08	0.00	-1.22
55	108 D.L.+ W.L(PUSH)+ W.L.(+Z)	-1.82	8.27	0.28	0.08	0.00	-0.13
55	109 D.L.+ W.L(PUSH)+ W.L.(-Z)	-1.81	7.66	-0.28	-0.08	0.00	-0.13
55	110 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)	2.71	-5.08	0.00	0.00	0.00	-1.42

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

55	111 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)	1.67	-5.08	0.00	0.00	0.00	-0.68
55	112 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.84	6.48	0.00	0.00	0.00	-0.60
55	113 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.88	6.48	0.00	0.00	0.00	0.14
55	114 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)	2.19	-4.85	0.21	0.06	0.00	-1.05
55	115 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)	2.19	-5.31	-0.21	-0.06	0.00	-1.05
55	116 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.84	6.48	0.00	0.00	0.00	-0.60
55	117 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.88	6.48	0.00	0.00	0.00	0.14
55	118 0.6 D.L.+ W.L(UPLIFT)+ 1W.L.(+X)	3.60	-8.26	0.00	0.00	0.00	-1.49
55	119 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.22	-8.26	0.00	0.00	0.00	-0.51
55	120 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.12	7.15	0.00	0.00	0.00	-0.40
55	121 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.51	7.16	0.00	0.00	0.00	0.58
55	122 0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.91	-7.96	0.28	0.08	0.00	-1.00
55	123 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.91	-8.57	-0.28	-0.08	0.00	-1.00

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

55	124 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.12	7.15	0.00	0.00	0.00	-0.40
55	125 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.51	7.16	0.00	0.00	0.00	0.58
56	101 D.L.	-0.04	3.01	-0.02	0.00	0.00	-0.74
56	102 D.L.+ W.L(UPLIFT)+ W.L.(+X)	3.78	-8.02	0.06	0.01	0.00	-1.66
56	103 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.37	-8.03	0.07	0.01	0.00	-0.86
56	104 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.27	8.53	-0.07	-0.01	0.00	-0.47
56	105 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.68	8.53	-0.07	-0.01	0.00	0.32
56	106 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	3.08	-8.35	1.48	0.31	0.00	-1.26
56	107 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	3.07	-7.70	-1.35	-0.30	0.00	-1.26
56	108 D.L.+ W.L(PUSH)+ W.L.(+Z)	-1.97	8.21	1.35	0.29	0.00	-0.08
56	109 D.L.+ W.L(PUSH)+ W.L.(-Z)	-1.98	8.86	-1.49	-0.31	0.00	-0.07
56	110 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)	2.83	-5.48	0.04	0.00	0.00	-1.37
56	111 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)	1.77	-5.48	0.05	0.01	0.00	-0.78
56	112 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.96	6.94	-0.06	-0.01	0.00	-0.49

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

56	113 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-2.02	6.94	-0.05	-0.01	0.00	0.11
56	114 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)	2.30	-5.72	1.11	0.23	0.00	-1.08
56	115 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)	2.30	-5.24	-1.02	-0.22	0.00	-1.07
56	116 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.96	6.94	-0.06	-0.01	0.00	-0.49
56	117 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-2.02	6.94	-0.05	-0.01	0.00	0.11
56	118 0.6 D.L.+ W.L(UPLIFT)+ 1W.L.(+X)	**3.79**	**-8.88**	0.07	0.01	0.00	-1.44
56	119 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)	2.38	-8.89	0.07	0.01	0.00	-0.65
56	120 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.26	7.67	-0.06	-0.01	0.00	-0.26
56	121 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.67	7.67	-0.06	-0.01	0.00	0.54
56	122 0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	3.09	**-9.21**	1.49	0.31	0.00	-1.05
56	123 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	3.08	-8.56	-1.35	-0.29	0.00	-1.04
56	124 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-1.26	7.67	-0.06	-0.01	0.00	-0.26
56	125 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-2.67	7.67	-0.06	-0.01	0.00	0.54
57	101 D.L.	0.03	2.26	-0.01	-0.01	0.00	-0.63

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

57	102 D.L.+ W.L(UPLIFT)+ W.L.(+X)	2.83	-5.83	0.02	0.03	0.00	-1.31
57	103 D.L.+ W.L(UPLIFT)+ W.L.(-X)	1.77	-5.83	0.02	0.03	0.00	-0.65
57	104 D.L.+ W.L(PUSH)+ W.L.(+X)	-0.88	6.28	-0.02	-0.03	0.00	-0.45
57	105 D.L.+ W.L(PUSH)+ W.L.(-X)	-1.93	6.28	-0.02	-0.03	0.00	0.21
57	106 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.32	-5.81	0.32	0.12	0.00	-1.01
57	107 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.27	-5.86	-0.27	-0.07	0.00	-0.95
57	108 D.L.+ W.L(PUSH)+ W.L.(+Z)	-1.38	6.31	0.27	0.07	0.00	-0.15
57	109 D.L.+ W.L(PUSH)+ W.L.(-Z)	-1.43	6.26	-0.32	-0.12	0.00	-0.09
57	110 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+X)	2.13	-3.97	0.02	0.02	0.00	-1.09
57	111 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-X)	1.33	-3.97	0.02	0.02	0.00	-0.60
57	112 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.65	5.12	-0.02	-0.02	0.00	-0.45
57	113 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.45	5.12	-0.02	-0.02	0.00	0.04
57	114 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(+Z)	1.75	-3.95	0.24	0.09	0.00	-0.87

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

57	115 D.L.+ 0.75*W.L(UPLIFT)+ 0.75*W.L.(-Z)	1.71	-3.99	-0.21	-0.05	0.00	-0.82
57	116 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(+X)	-0.65	5.12	-0.02	-0.02	0.00	-0.45
57	117 D.L.+ 0.75*W.L(PUSH)+ 0.75*W.L.(-X)	-1.45	5.12	-0.02	-0.02	0.00	0.04
57	118 0.6 D.L.+ W.L(UPLIFT)+ 1W.L.(+X)	2.82	-6.48	0.02	0.03	0.00	-1.13
57	119 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-X)	1.76	-6.48	0.02	0.03	0.00	-0.47
57	120 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-0.89	5.64	-0.02	-0.02	0.00	-0.27
57	121 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-1.94	5.64	-0.02	-0.02	0.00	0.39
57	122 0.6 D.L.+ W.L(UPLIFT)+ W.L.(+Z)	2.31	-6.46	0.32	0.12	0.00	-0.83
57	123 0.6 D.L.+ W.L(UPLIFT)+ W.L.(-Z)	2.26	-6.50	-0.27	-0.06	0.00	-0.77
57	124 0.6 D.L.+ W.L(PUSH)+ W.L.(+X)	-0.89	5.64	-0.02	-0.02	0.00	-0.27
57	125 0.6 D.L.+ W.L(PUSH)+ W.L.(-X)	-1.94	5.64	-0.02	-0.02	0.00	0.39
51	101 D.L.	0.03	2.26	0.01	0.01	0.00	-0.63

5.5 Foundation Design Forces

Uplift = 8.8 kN

Compression = 9.21 kN

Shear = 3.8 kN

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Moment = 1.71 kN-m

6. DESIGN OF MMS CONCRETE PILE FOUNDATION.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. DESIGN OF MMS CONCRETE PILE FOUNDATION.

SN	Item Description	Symbol		Value	Unit
A	**Foundation Reactions**				
	Uplift forces on the foundation	Fu	=	**8.80**	KN
	Downward forces on the foundation	Fy	=	**9.21**	KN
	Shear value from super-structure	Q	=	**3.80**	KN
	Max Moment from super-structure	Qm	=	**1.90**	KN-m
B	**Pile Specification**				
	Length of pile below ground Lf	Lf	=	1400	mm
	Diameter of pile	D	=	300	mm
	Density of concrete	ρc	=	2400	kg/m^3
	Density of soil	ρs	=	19	kN/m^3
	Angle of friction	φ	=	35	degrees
	Unsupported length above ground	e	=	50	mm
	Strength of concrete (SBC- 303 _16.1.1)	Fck	=	25	N/mm^2
	Cohesion	C	=	0	N/mm^2
	Reduction factor	α	=	1	
	Standard Penetration Test(N)	N	=	40	Blows/ 30cm
	Relative Density	D_R	=	75	%
C	**Calculation of Net Allowable Load after considering the factor of safety for soil**				
	Weight of pile $\mathrm{Wt\ pile} = (\frac{\pi}{4}.\mathrm{D}^2).(\mathrm{H+e}).\rho_c$		=	2.41	KN
	Net uplift load on column $\mathrm{F_{net}} = \mathrm{F_{uplift}} - \mathrm{Wt_{pile}}$		=	6.39	KN
	Earth pressure Coefficient (Table 8- BS 8004-CFA PILE)	Ks	=	0.90	
	Surface area of the pit $\mathrm{A_{pit}} = (\pi.\mathrm{D_{pile}}).(\mathrm{H})$		=	1.32	m^2
	Resisting force due to skin friction δ = $K_δ$* φ ($K_δ$ = 1) (Table 8- BS 8004-CFA PILE)			35.00	KN
	Resisting force due to skin friction (EQ-4.8) $\mathrm{P_1} = 0.5 * (\mathrm{K_s}).(\rho_{soil}).(\mathrm{H}).(\mathrm{A_{pit}}).\tan(\delta)$			11.06	KN
	Partial Factor(Table A.7-Eurocode-7- R1 Set)	$γ_{Rd}$	=	1.25	
	Resisting loads for soil, $\mathrm{P_s} = (\mathrm{P_1})/(\mathrm{FOS})$		=	8.85	KN
	Resisting Load is greater than Net uplift force			***HENCE SAFE***	***0.73***
D	**Calculation of Compression or Bearing Resistance**				
	The Ultimate Load Capacity of Pile	Q_u	=	$Q_{u,E}+Q_{u,F}$	
	Load resisting due to friction $Qu,F = P_1$		=	11.06	KN
	Load resisting due to end bearing $Qu,E = P_2$		=		KN
	Bearing Factor	N_q	=	30	
	End bearing resistance in granular soils (EQ-4.8) $\mathrm{P2} = \mathrm{Ap}.(ɣ.\mathrm{He}.\mathrm{Nq})$		=	56.407	KN
	The Ultimate Load Capacity of Pile	Q_u	=	67.47	KN
	Partial Factor(Table A.7-Eurocode-7)	$γ_{Rd}$	=	1.25	
	Resisting loads for soil, $\mathrm{P_s} = (\mathrm{P_1} + \mathrm{P_2})/(\mathrm{FOS})$		=	53.97	KN
	Resisting Load is greater than bearing force			***HENCE SAFE***	**0.22**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

E	**Determination of type of Pile**				
	Length of the pile	L = H + e	=	1.45	m
	Modulus of subgrade reaction(Cohesionless)	ηh	=	9	MN/m^3
	Modulus of subgrade reaction(Cohesive)	K	=	1	MN/m^3
	Modulus of elasticity of concrete	E	=	25000	Mpa
	Moment of Inertia of pile	$I = \frac{\pi}{4}.(\frac{D_{pile}}{2})^4$	=	397607820	mm^4
	Stiffness Factor (Cohesionless)	$T = (\frac{EI}{\eta h})^{(\frac{1}{5})}$	=	1.020	m
	Stiffness Factor (Cohesive Soil)	$R = (\frac{EI}{KB})^{(\frac{1}{4})}$	=	2.399	m
	Cohesionless Soli selected, Stiffness Factor selected is T				
	SHORT PILE				
F	**Check for Ultimate Soil Capacity (Brom's Method)**				
	Passive Earth pressure co-efficient	$K_p = \frac{1+\sin(\varphi)}{1-\sin(\varphi)}$	=	3.69	
	Effective soil density,	γ = ρ(soil)	=	19.00	KN/m^3
	Unsupported length above ground	e	=	50	mm
	Length of pile below ground	H	=	1400	mm
	Diameter of pile	D	=	300	mm
	Check for Lateral Shear:				
		H/B	=	4.7	
		e/B	=	0.2	
		Horizontal shear factor(H_f)		6.26	
	Ultimate lateral resistance due to cohesion(Figure No-8.4(a))	$Q_{uc} = H_f.C.D^2$	=	0	KN
	Ultimate Resistance under friction, (EQ No-8.10)	$Q_{uf} = \frac{0.5.\gamma.H^3.D.K_p}{e+H}$	=	19.9024991	KN
	Ultimate Load Capacity	$Q_u = Q_{uc} + Q_{uf}$	=	19.90	KN
	Factor of Safety	FOS	=	2.000	
	Allowable Load Capacity	Q_{all}	=	9.95	KN
	As per the staad report, Shear value from super-structure	Q	=	3.8	KN
	Allowable Load Capacity is greater than Net Shear from Superstructure		***HENCE SAFE***		**0.39**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

G	**Check for Bending Moment:**				
	Resistance to moment due to cohesion, (EQ No-8.1)	$f=(Q_{uc}/9.C.B)$	=	0	m
	Value of(EQ No-8.3)	$M_{maxc} = [Q_{uc}.(e + 1.5.D + 0.5.f)]$	=	0	KN-m
	Resistance to moment due to friction(EQ No-8.11)	$X_{0f} = 0.82.\sqrt{\frac{Q_{uf}}{\gamma.D.K_p}}$	=	0.79763997	m
	Value of (EQ No-8.12)	$M_{maxf} = Q_{uf}.(e + 0.667.X_{0f})$	=	11.584	KN-m
	Overall Resistance	$M_{max} = M_{maxc} + M_{maxf}$	=	11.584	KN-m
	Factor of safety	FOS	=	2.000	
	Allowable Max Moment	$M = \frac{M_{max}}{FOS}$	=	5.79	KN-m
	Max Moment from super-structure	Qm	=	1.9	KN-m
	Allowable Max Moment is greater than Max Moment from Superstructure		***HENCE SAFE***		**0.33**
H	**Lateral deflection check - As per broms Methd**				
	Cohesionless Soil				
	Allowable lateral deflection Assumed (SBC-303_CL14.9.3)		=	12.5	mm
	Dia of pile d		=	0.3	m
	unsupported length of pile e = L1		=	0.05	m
	Length of pile below ground level L		=	1.4	m
	Horizontal load from the analysis		=	3.80	KN
	Grade of concrete		=	25	N/mm^2
	Youngs modulus of concrete = 5000*fck^1/2		=	25000	MN/m^2
	Youngs modulus of concrete MN/m^2		=	25000	N/mm^2
	Moment of inertia of pile cross section in mm^4 = ∏*d^4/64		=	0.00039761	m^4
	Coefficient of modular variation ηh in MN/m^3 ((Figure 6.20)		=	9	MN/m^3
	To find dimensionless depth of embedment (EQ No-6.47)	$\eta = (\frac{\eta h}{EI})^{(\frac{1}{5})}$	=	0.98032373	
	ηL	ηL	=	1.37	
				Short Pile	
	Deflection of pile embedded in cohesionless soil(Y_o) (EQ No-6.48)	$Yo = \frac{18H * (1 + \frac{1.33e}{L})}{L^2 * \eta h}$	=	**4.06**	mm
	Allowable Deflection is geater than acctual deflection		***HENCE SAFE***		**0.33**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7. DESIGN OF MMS FOUNDATION WITH STAAD PRO.

The foundation is modelled with square member element and base reactions are applied at top.Lateral springs are provided along the height of the foundation in contact with soil.Vertical and rotational springs are applied at the base level.

Charaterisitics of Pole pedestal :

Dimensions at the base :

Width : 30 (cm)

Height Real : 30 (cm)

Height Effective under compression : 30 (cm) (Refer 8.3 Clause)

Pole pedestal Height :

Projection above ground : 5 (cm)

Embedded length : 150 (cm)

Applied loads on the pole pedestal :

Selfweight applied directly in STAAD

Forces from super structure.

Vertical Load , N : 0.92 (Ton)

Horizontal Load , Q : 0.380 Ton

Moment Load , M : 0.190 Ton-m

7.1. Soil Spring Calculation

Co-efficient of horizontal modulus of subgrade reaction = 3220 Ton/m3 (As per table-9 of Geo-technical report)

K lateral at Z= 1.2 m 3220 Ton/m3 K lateral at Z= 1.5 m 3890 Ton/m3

Horizontal modulus of subgrade reaction at 1.2m is considered for the design

Lateral Area #1 = 0.0420

Lateral Area #2 = 0.0210

Vertical Position	Coordinate Z	Soil	K lateral (Ton/m3)	Spring	K lateral (Ton/m)
0	0.000		0		0.000
0.14	0.140		322		13.524
0.28	0.280		644		27.048
0.42	0.420		966		40.572
0.56	0.560		1288		54.096
0.7	0.700		1610		67.620
0.84	0.840		1932		81.144
0.98	0.980		2254		94.668
1.12	1.120		2576		108.192
1.26	1.260		2898		121.716
1.4	1.400		3220		67.620

Co-efficient of vertical modulus of subgrade reaction = 4920 Ton/m3 (As per table-9 of Geo-technical report)

Vertical spring constant at the node at base level considering effective area = (0.3x0.3x4920) = 442.8 Ton/m

Co-efficient of rotational modulus of subgrade reaction = 4920 x 2=9840 Ton/m3

(Considered to be 2 times of vertical modulus of subgrade reaction value i.e 4920 Ton/m3 as per table -9 of Geo-technical report)

Moment of Inertia of effective area under compression = $(0.3x0.3^3)/12$ = 0.000675 m4

Rotational spring = 9840 x 0.000675 = 6.642 Ton-m / degree

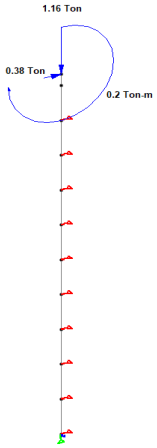
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

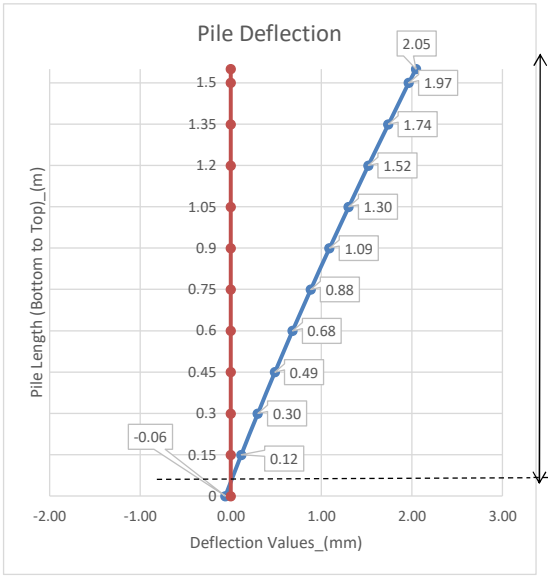
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.2. Pile Deflection Graph

Vertical Load , N : 1.16 (Ton)
Horizontal Load , Q : 0.38 Ton
Moment Load , M : 0.2 Ton-m



STAAD MODEL



1350 mm

100mm
50 mm

DEFLECTION DIAGRAM

Max deflction at top = 2.19 mm

Allowable deflection at top of pipe
=12.5 mm (Refer SBC-303_CL14.9.3)
HENCE , OK

Deflection = 0.12mm

Tanα = 2.05/(1350) = 0.0016 < 0.01
As per Sulzberger guidelines for slope gradient tanα <0.01, soil has elastic behaviour)

HENCE O.K.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

7.3. Calculation of Effective Compression Height at Base

Soil Properties :

Lateral ballast coefficient at baseline; C_L : 3.22 (Kg/cm3) (As per table-12 of Geo-technical report)

Base line rotational ballast coefficient; C_ϕ : 9.84 (Kg/cm3) (Considered to be 2 times of vertical modulus of subgrade reaction value i.e 2.82 Kg/cm3 as per table -12 of Geo-technical report)

Charaterisitics of Pole pedestal :

Dimensions at the base :

Width : 30 (cm)

Height Real : 30 (cm)

Height Effective under compression : 30 (cm)

Pole pedestal Height :

Projection above ground : 0 (cm)

Embedded length : 150 (cm)

Applied loads on the pole pedestal :

Self weight of pole pedestal = 0.7853x0.3x0.3x1.55x2.3 = 0.24 (Ton) (Unit wt of concrete= 2.3 Ton/m3)

Vertical Load , N : 0.92 (Ton)

Vertical Load , N : 0.25 + 1.17 = 1.16 (Ton)

Horizontal Load , Q : 0.38 (Ton)

Bending Moment, M : 0.19 (Ton-m) (Unfactored moment)

Ys = Upper displacement = 0.64 cm ?

Yi = Lower displacement = -0.24 cm

Angle of rotation of pole foundation = 0.01

Moment at stack base 0.039 (Ton-m)

Pressure at base

Trapezoidal diagram

max. Pressure = 2.151 kg/cm2

min. Pressure = 0.420 kg/cm2 > 0 Hence O.K.

Calculation area :

Determination of Displacements

Moment of Inertia	67500.00	Effective area (Af) =	900 Only compressed section
Rigidity Lateral	96.6	Section Modulus (Wf) =	4500.00 Only compressed section
Stiffness	664200.00		

Stiffness Matrix

185553	176697	Ys	76000
1	2	Yi	0.1573

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8. SOIL DESIGN PARAMETERS FROM GEOTECH REPORT

Geotechnical Design Parameters – Solar Plant & Workshop Area (BH-05)

Soil Type	Depth (m)		Design SPT Value	Design RQD Value	Bulk Density "γ_b"	Undrained Cohesion "C_u"	Drained Cohesion "C"	Angle of Internal Friction "φ"	Modulus of Elasticity "E_s"
	From	To	N_{60}	(%)	(kN/m^3)	(kPa)	(kPa)	(degree)	(kPa)
SAND with Silt and Gravel / GRAVEL with Silt and Sand	0.00	0.75	30	0	18	-	0	33	30,000
	0.75	15.0	100	0	20	-	0	37	100,000

As the foundation type is bored cast insitu mono concrete pile, the foundation is designed for the single layer of soil, by assuming the following average design parameters

SPT Value = 40 Blows/30cm

Bulk Density = 19 kN/m^3

Angle of Internal Friction = 19 kN/m^3

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

[1] Structural Loading and Forces ; SBC_Code_301_2007

[2] Saudi-Building-Code-Structural-Soil-and-Foundations:SBC_Code_303_2007

[3] Civil, Structural and Architectural Design Criteria; MD-216-1000-EG-ST-CRT-1001

[4] Pile Design and Construction Practice By M.J.Tomlinson

[5] The Foundation Engineering Handbook, By Manjriker Gunaratne

[6] BS 8004 : Code of Practice for Foundations

[7] Eurocode 7: Geotechnical design - Part 1: General rules

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 STAAD INPUT

```
**************************************************
*                                                *
*      STAAD.Pro V8i SELECTseries5               *
*      Version  20.07.10.41                      *
*      Proprietary Program of                    *
*      Bentley Systems, Inc.                     *
*      Date=    AUG 29, 2020                     *
*      Time=    12:20: 1                         *
*                                                *
*    USER ID: LNT ECC DIV                        *
**************************************************
```

1. STAAD PLANE
   INPUT FILE: MMS Foundation _v01.STD
2. START JOB INFORMATION
3. ENGINEER DATE 8.11.2019
4. JOB NAME POST FOOUNDATION
5. ENGINEER NAME SOR
6. CHECKER NAME SJM
7. APPROVED NAME AG
8. JOB REV 0
9. JOB CLIENT MAADEN
10. JOB NO O19045
11. END JOB INFORMATION
12. INPUT WIDTH 79

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

13. UNIT METER MTON
14. JOINT COORDINATES
15. 1 0 4.47 0; 2 0 4.61 0; 3 0 4.75 0; 4 0 4.89 0; 5 0 5.03 0; 6 0 5.17 0
16. 7 0 5.31 0; 8 0 5.45 0; 9 0 5.59 0; 10 0 5.73 0; 11 0 5.87 0; 12 0 5.92 0
17. MEMBER INCIDENCES
18. 1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6; 6 6 7; 7 7 8; 8 8 9; 9 9 10; 10 10 11
19. 11 11 12
20. **********
21. DEFINE MATERIAL START
22. ISOTROPIC CONCRETE
23. E 2.21467E+006
24. POISSON 0.17
25. DENSITY 2.3
26. ALPHA 1E-005
27. DAMP 0.05
28. TYPE CONCRETE
29. STRENGTH FCU 2812.28
30. ISOTROPIC STEEL
31. E 2.09042E+007
32. POISSON 0.3
33. DENSITY 7.83341
34. ALPHA 1.2E-005
35. DAMP 0.03
36. TYPE STEEL
37. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
38. END DEFINE MATERIAL

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

STAAD PLANE -- PAGE NO. 2

39. MEMBER PROPERTY AMERICAN
40. 1 TO 11 PRIS YD 0.3 ZD 0.3
41. CONSTANTS
42. MATERIAL CONCRETE ALL
43. **********
44. SUPPORTS
45. 10 FIXED BUT FY FZ MX MY MZ KFX 13.52
46. 9 FIXED BUT FY FZ MX MY MZ KFX 27.048
47. 8 FIXED BUT FY FZ MX MY MZ KFX 40.572
48. 7 FIXED BUT FY FZ MX MY MZ KFX 54.096
49. 6 FIXED BUT FY FZ MX MY MZ KFX 67.62
50. 5 FIXED BUT FY FZ MX MY MZ KFX 81.144
51. 4 FIXED BUT FY FZ MX MY MZ KFX 94.668
52. 3 FIXED BUT FY FZ MX MY MZ KFX 108.192
53. 2 FIXED BUT FY FZ MX MY MZ KFX 121.716
54. 1 FIXED BUT FZ MX MY KFX 67.62 KFY 442.8 KMZ 6.642
55. LOAD 1 LOADTYPE NONE TITLE MMS STRUCTURE LOAD
56. JOINT LOAD
57. 12 FX 0.38 FY -1.16 MZ -0.2
58. PERFORM ANALYSIS

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	12	NUMBER OF MEMBERS	11
NUMBER OF PLATES	0	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	10

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH = 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 1,
TOTAL DEGREES OF FREEDOM = 36
SIZE OF STIFFNESS MATRIX = 1
DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 550923.5 MB

61. FINISH

********** END OF THE STAAD.Pro RUN **********

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 2 FORMULA REFERENCE PAGES

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

8

Design of Laterally Loaded Piles

Manjriker Gunaratne

CONTENTS

8.1 Introduction 327
8.2 Lateral Load Capacity Based on Strength 328
8.2.1 Ultimate Lateral Resistance of Piles 328
8.2.1.1 Piles in Homogeneous Cohesive Soils 328
8.2.1.2 Piles in Cohesionless Soils 335
8.3 Lateral Load Capacity Based on Deflections 339
8.3.1 Linear Elastic Method 339
8.3.1.1 Free-Headed Piles 339
8.3.1.2 Fixed-Headed Piles 341
8.3.2 Nonlinear Methods 343
8.3.2.1 Stiffness Matrix Analysis Method 343
8.3.2.2 Lateral Pressure–Deflection (p–y) Method of Analysis 348
8.3.2.3 Synthesis of p–y Curves Based on Pile Instrumentation 350
8.4 Lateral Load Capacity of Pile Groups 355
8.5 Load and Resistance Factor Design for Laterally Loaded Piles 356
8.6 Effect of Pile Jetting on the Lateral Load Capacity 356
8.7 Effect of Preaugering on the Lateral Load Capacity 360
References 361

8.1 Introduction

Single piles such as sign-posts and lamp-posts and pile groups that support bridge piers and offshore construction operations are constantly subjected to significant natural lateral loads (such as wind loads and wave actions) (Figure 8.1). Lateral loads can be also introduced on piles due to artificial causes like ship impacts. Therefore, the lateral load capacity is certainly a significant attribute in the design of piles under certain construction situations.

Unlike in the case of axial load capacity, the lateral load capacity must be determined by considering two different failure mechanisms: (1) structural failure of the pile due to yielding of pile material or shear failure of the confining soil due to yielding of soil, and (2) pile becoming dysfunctional due to excessive lateral deflections. Although passive failure of the confining soil is a potential failure mode, such failure occurs only at relatively large deflections which generally exceed the tolerable movements.

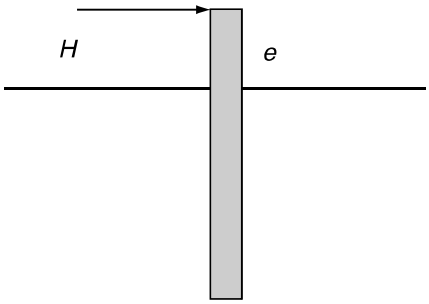


FIGURE 8.1
Laterally loaded pile.

One realizes that ''short'' piles embedded in relatively stiffer ground would possibly fail due to yielding of the soil while ''long'' piles embedded in relatively softer ground would produce excessive deflections. In view of the above conditions, this chapter is organized to analyze separately, the two distinct issues presented above. Hence the discussion will deal with two main issues: (1) lateral pile capacity from strength considerations, and (2) lateral pile capacity based on deflection limitations.

On the other hand, piles subjected to both axial and lateral loading must be designed for structural resistance of the piles as beam-columns.

8.2 Lateral Load Capacity Based on Strength

8.2.1 Ultimate Lateral Resistance of Piles

Broms (1964a,b) produced simplified solutions for the ultimate lateral load capacity of piles by considering both the ultimate strength of the bearing ground and the yield stress of the pile material. For simplicity, the Broms (1964a,b) solutions are presented separately for different soil types, namely, cohesive soils and cohesionless soils.

8.2.1.1 Piles in Homogeneous Cohesive Soils

When a pile is founded in a predominantly fine-grained soil, the most critical design case is the case where soil is in an undrained situation. The maximum load that can be applied on the pile depends on the the following factors:

1. Fixity conditions at the top (i.e., free piles or fixed piles). Most single piles can be considered as free piles under lateral loading whereas piles clustered in a group by a pile cap must be analyzed as fixed piles.
2. Relative stiffness of the pile compared to the surrounding soil. If the deformation conditions are such that the soil yields before the pile material then the pile is classified as a ''short'' pile. Similarly, if the pile material yields first, then the pile is considered a ''long'' pile.

8.2.1.1.1 Unrestrained or Free-Head Piles

Figure 8.2 and Figure 8.3 illustrate the respective failure mechanisms that Broms (1964a,b) assumed for ''short'' and ''long'' piles, respectively.

The ultimate lateral resistance P_u can be directly determined from Figure 8.4(a) and (b) based on the geometrical properties and the undrained soil strength. For short piles, M_{max}, g, P_u, and f can be determined from Equations (8.1) to (8.4).

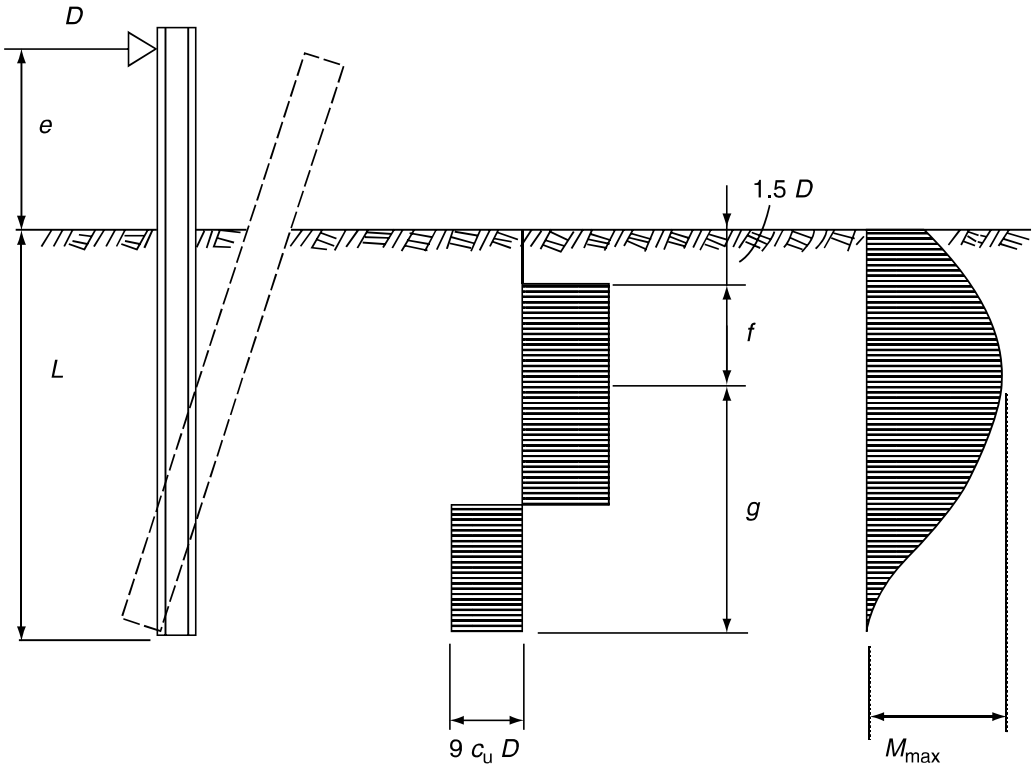


FIGURE 8.2
Deflection, soil reaction, and bending moment distributions for laterally loaded short piles in cohesive soil. (From Broms, B., 1964a, *J. Soil Mech. Found. Div., ASCE*, 90(SM3):27–56. With permission.)

Since the shear force is zero at the location of maximum moment, from the area of the soil reaction plot (Figure 8.2) one obtains

$$f = \frac{P_u}{9c_u D} \tag{8.1}$$

Similarly, by taking the first moments of Figure 8.2 about the yield point

$$M_{max} = 2.25Dg^2c_u \tag{8.2}$$

$$M_{max} = H_u(e + 1.5D + 0.5f) \tag{8.3}$$

For the total length of the pile,

$$L = g + 1.5D + f \tag{8.4}$$

8.2.1.1.2 Restrained or Fixed-Head Piles

According to the Broms (1964a) formulations, restrained piles can reach their ultimate capacity through three separate mechanisms giving rise to (1) short piles, (2) long piles, and (3) intermediate piles. These failure mechanisms assumed by Broms (1964a) for restrained piles are illustrated in Figure 8.5(a)–(c). The assumption that leads to the analytical solutions is that the moment generated on the pile top can be provided by the pile cap to restrain the pile with the boundary condition at the top (i.e., no rotation).

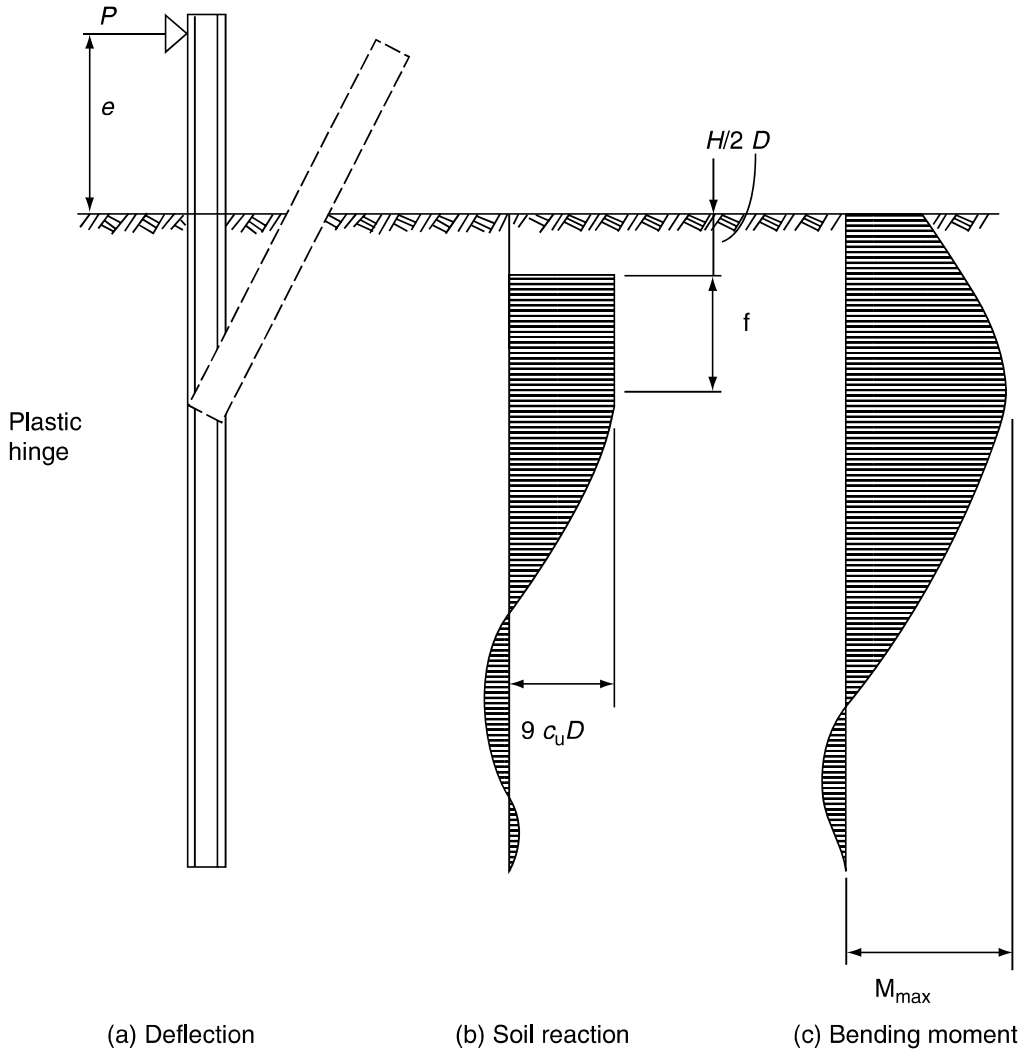


FIGURE 8.3
Failure mechanism for laterally loaded long piles in cohesive soil. (From Broms, B., 1964a, *J. Soil Mech. Found. Div., ASCE*, 90(SM3):27–56. With permission.)

The ultimate lateral load, P_u, of short piles can be directly obtained from Figure 8.4(a). The reader would notice that this condition is presented through a single curve in Figure 8.4(a) due to the insignificance of the e parameter. M_{max} and KP_u can also be determined using the following equations:

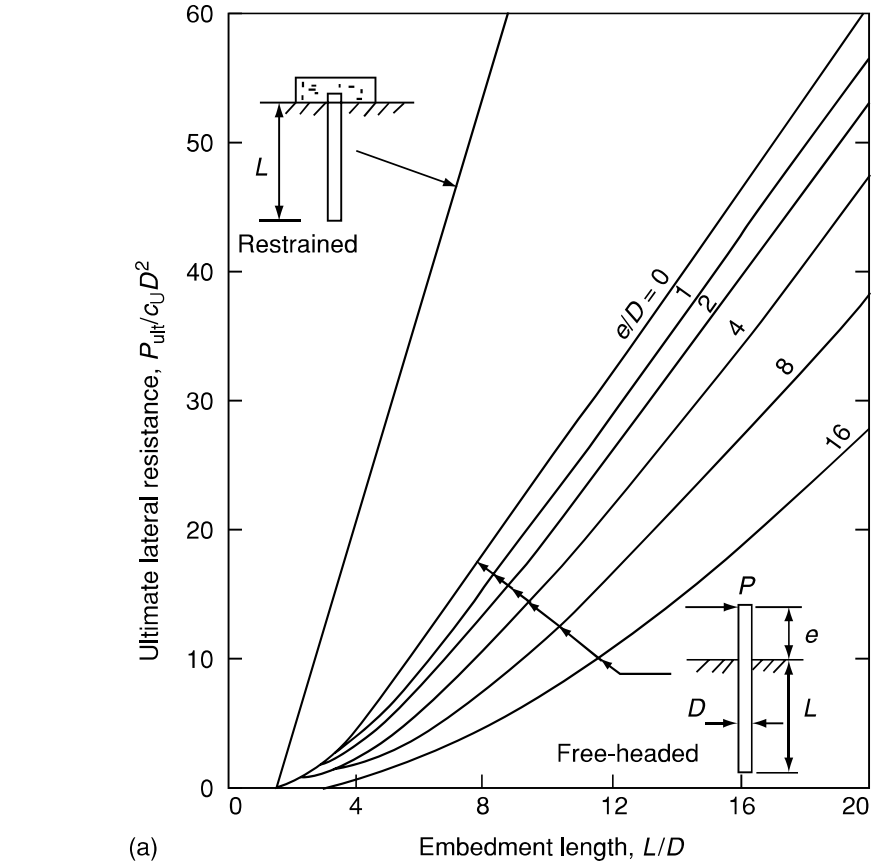
$$P_u = 9c_u D(L - 1.5D) \tag{8.5}$$

$$M_{max} = P_u(0.5L + 0.75D) \tag{8.6}$$

For long piles, the ultimate lateral load, P_u, can be found from Figure 8.4(b). Then, the following equations can be used to determine f and hence the location of pile yielding:

$$P_u = \frac{2M_y}{(1.5d + 0.5f)} \tag{8.7}$$

On the other hand, for ''intermediate'' piles where yielding occurs at the top (Figure 8.5b), the basic shear moment and total length consideration in Equations (8.1), (8.4), and (8.8) can be used to obtain P_u:



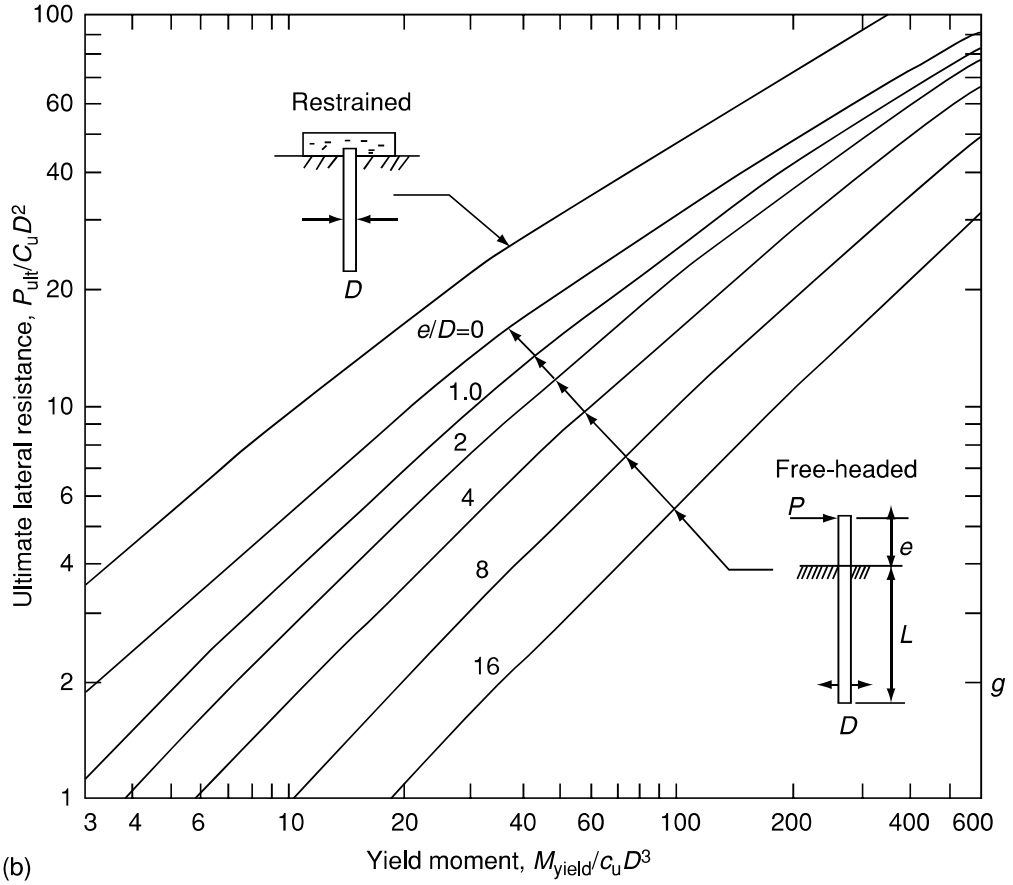
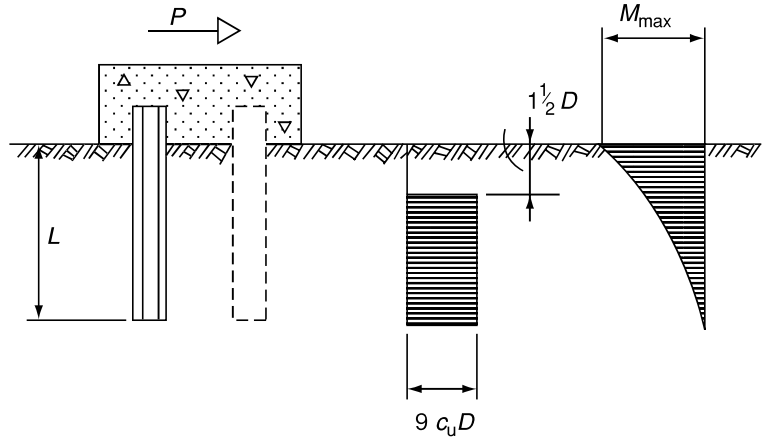
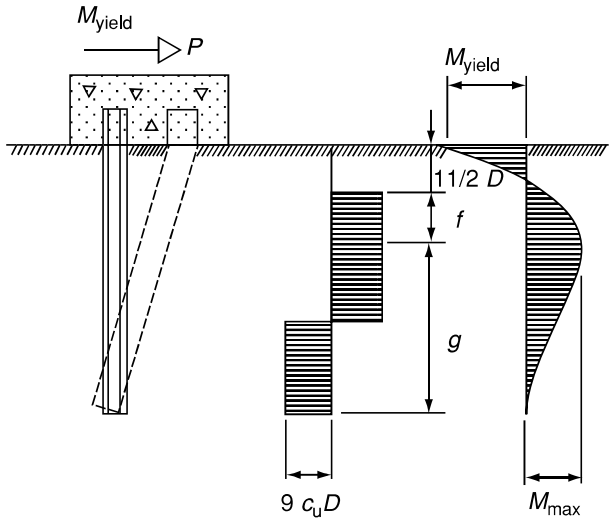


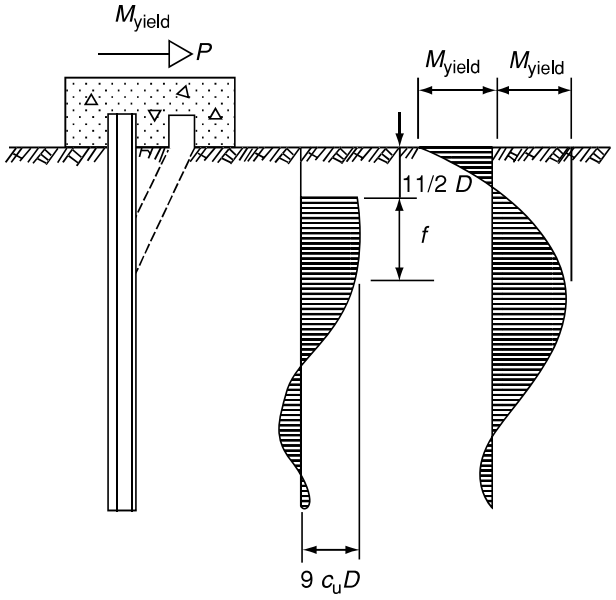
FIGURE 8.4
Ultimate lateral resistance of piles in cohesive soils: (a) short piles and (b) long piles. (From Broms, B., 1964a, *J. Soil Mech. Found. Div., ASCE*, 90(SM3):27–56. With permission.)



(a) (a) Deflection (b) Soil reaction (C) Bending moment



(b) (a) Deflection (b) Soil reaction (c) Bending moment



(c) (a) Deflection (b) Soil reaction (c) Bending moment

FIGURE 8.5
Failure mechanisms for laterally loaded restrained piles in cohesive soils: (a) short piles, (b) intermediate piles and (c) long piles. (From Broms, B., 1964a, *J. Soil Mech. Found. Div., ASCE*, 90(SM3):27–56. With permission.)

on reductions in resistance with time for loading tests made at a rapid rate of application on piles in stiff clays are as follows.

Type of pile	Type of clay	Change in resistance	Reference
Driven precast concrete	London	Decrease of 10 to 20% at 9 months over first test at 1 month	Meyerhof and Murdock(4.11)
Driven precast concrete	Aarhus (Septarian)	Decrease of 10 to 20% at 3 months over first test at 1 month	Ballisager(4.19)
Driven steel tube	London	Decrease of 4 to 25% at 1 year over first test at 1 month	Tomlinson(4.5)

It is important to note that the same pile was tested twice to give the reduction shown above. Loading tests on stiff clays often yield load-settlement curves of the shape shown in Figure 11.13b (Section 11.4.2). Thus the second test made after a time interval may merely reflect the lower 'long-strain' skin friction which has not recovered to the original peak value at the time of the second test. From the above data it is concluded that the fairly small changes in pile resistance for periods of up to one year are of little significance compared with other uncertain effects. An increase should be allowed only in the case of soft clays sensitive to remoulding.

4.3 Piles in cohesionless soil

4.3.1 General

The classic formulae for calculating the resistance of piles in cohesionless soils follow the same form as equation 4.1. Expressed in the parameters of a cohesionless soil (c_u=0), the total pile resistance is given by the expression

$$Q_p = N_q \sigma'_{vo} A_b + \frac{1}{2} K_s \sigma'_{vo} \tan \delta A_s \qquad \ldots(4.8)$$

where σ'_{vo} is the effective overburden pressure at pile base level, N_q is the bearing capacity factor, A_b is the area of the base of the pile, K_s is a coefficient of horizontal soil stress which depends on the relative density and state of consolidation of the soil, the volume displacement of the pile, the material of the pile, and its shape, δ is the characteristic or average value of the angle of friction between pile and soil, depending on whether or not the Eurocode rules are followed, and A_s is the area of shaft in contact with the soil. The factors N_q and K_s are empirical and based on correlations with static loading tests, δ is obtained from empirical correlations with field tests. Hence when using Eurocode 7 the characteristic values of Q_{bk} and Q_{sk} should be factored to obtain the design bearing capacity Q. Table 2b of the code recommends a partial material factor of 1.2 to 1.25 for the parameter tan ϕ. The same factor can be used for *tan* δ in equation 4.8.

The factor N_q depends on the ratio of the depth of penetration of the pile to its diameter and on the angle of shearing resistance ϕ of the soil. The latter is normally obtained from the results on tests made in-situ (see 11.1.4). The relationship between the standard penetration resistance N and ϕ, as established by Peck, Hanson and Thornburn(4.20), and between the limiting static cone resistance, q_c and ϕ as established by Durgunoglu and Mitchell(4.21), are shown in Figures 4.13, and 4.14 respectively.

From tests made on instrumented full scale piles, Vesic(4.22) showed that the increase of base resistance with increasing depth was not linear as might be implied from equation 4.8, but that *rate* of increase actually decreased with increasing depth. For practical design purposes it has been assumed that the increase is linear for pile penetrations of between 10 and 20 diameters, and that below these depths the unit base resistance has been assumed to be at a constant value. This simple design approach was adequate for ordinary foundation work where the penetration depths of closed end piles were not usually much greater than 10 to 20 diameters. At these depths practical refusal was usually met with driving piles into medium dense to dense cohesionless soils.

However, the use of piled foundations for offshore petroleum production platforms has necessitated driving hollow cylindrical piles with open ends to very great depths below the sea bed to obtain resistance in skin friction to uplift loading. The assumption of a constant unit base resistance below a penetration depth of 10 to 20 diameters has been shown to be over-

conservative. It can be demonstrated by theoretical analysis, and proved by field experience, that the base resistance does not remain constant or reduce with depth. However, the *rate of increase* does increase with depth in a soil deposit of uniform density.

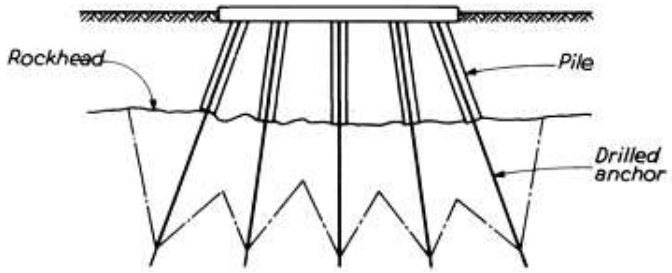


Fig. 6.16 Splaying anchors in group to increase uplift resistance

of inclination θ to a horizontal rock surface can be performed with the aid of the curve for V_c/L^3 in Figure 6.17a. The effect of overlapping cones of rock in groups of vertical or raking anchors can be calculated by reference to Figures 6.17a and 6.17b. These charts enable the overlapping volumes ΔV_m and ΔV_n to be calculated for a group of anchors arranged on a rectangular grid. They are not applicable to a diagonal (*i.e.* 'staggered') pattern. All the anchors in the group are assumed to be arranged at the same angle of inclination to the horizontal and the charts are based on a cone with a 30° half-angle. The charts are not valid if the sum of $(P/n)^2$ and $(S/m)^2$ is less than 4 when composite overlapping occurs. In such a case the total volume acting against uplift should be estimated from the geometry of the system.

Because of the various uncertainties in the design of rock anchors as described above, it is evident that it is desirable to adopt stressed anchors with every anchor individually stressed and hence checked for pull-out resistance at a proof load of 1.5 times the working load. However, it should be noted that the technique of stressing anchors by jacking against the reaction provided by the pile does not check the pull-out resistance of the cone of rock: this is clear from Figure 6.10. The resistance offered by the mass or rock can be tested only by providing a reaction beam with bearers sited beyond the influence of the conjectural rock cone. Tests of this description are very expensive to perform and it is usual to avoid them by adopting conservative assumptions for the dimensions of the cone, and applying a safety factor to the calculated weight if required.

6.3 Single vertical piles subjected to lateral loads

The ultimate resistance of a vertical pile to a lateral load and the deflection of the pile as the load builds up to its ultimate value are complex matters involving the interaction between a semi-rigid structural element and the soil, which deforms partly elastically and partly plastically. Taking the case of a vertical pile unrestrained at the head, the lateral loading on the pile head is initially carried by the soil close to the ground surface. At a low loading the soil compresses elastically but the movement is sufficient to transfer some pressure from the pile to the soil at a greater depth. At a further stage of loading the soil yields plastically and transfers its load to greater depths. A short rigid pile unrestrained at the top and having a length to width ratio of less than 10 to 12 (Figure 6.18a) tends to rotate and passive resistance develops above the toe on the opposite face to add to the resistance of the soil near the ground surface. Eventually the rigid pile will fail by rotation when the passive resistance of the soil at the head and toe are exceeded. The short rigid pile restrained at the head by a cap or bracing will fail by translation in a similar manner to an anchor block which fails to restrain the movement of a retaining wall transmitted through a horizontal tied rod (Figure 6.18b).

The failure mechanism of an infinitely long pile is different. The passive resistance of the lower part of the pile is infinite, and thus rotation of the pile cannot occur, the lower part remaining vertical while the upper part deforms to a shape shown in Figure 6.19a. Failure takes place when the pile fractures at the point of maximum bending moment, and for the purpose of analysis a plastic hinge capable of transmitting shear is assumed to develop at the point of fracture. In the case of a long pile restrained at the head, high bending stresses develop at the point of restraint, *e.g.* just beneath the pile cap, and the pile may fracture at this point (Figure 6.19b).

The pile head may move horizontally over an appreciable distance before rotation or failure of the pile occurs, to such an extent that the movement of the structure supported by the pile or pile group exceeds tolerable limits. Therefore, having calculated the ultimate load and divided it by the appropriate safety factor, it is still necessary to check that the permissible deflection of the pile is not exceeded.

Quite extensive research has been undertaken on the resistance of piles to lateral loading but this research has not yielded any simple design method which can be universally applied to any soil or type of pile. There are many inter-related factors. The dominant one is the pile stiffness, which influences

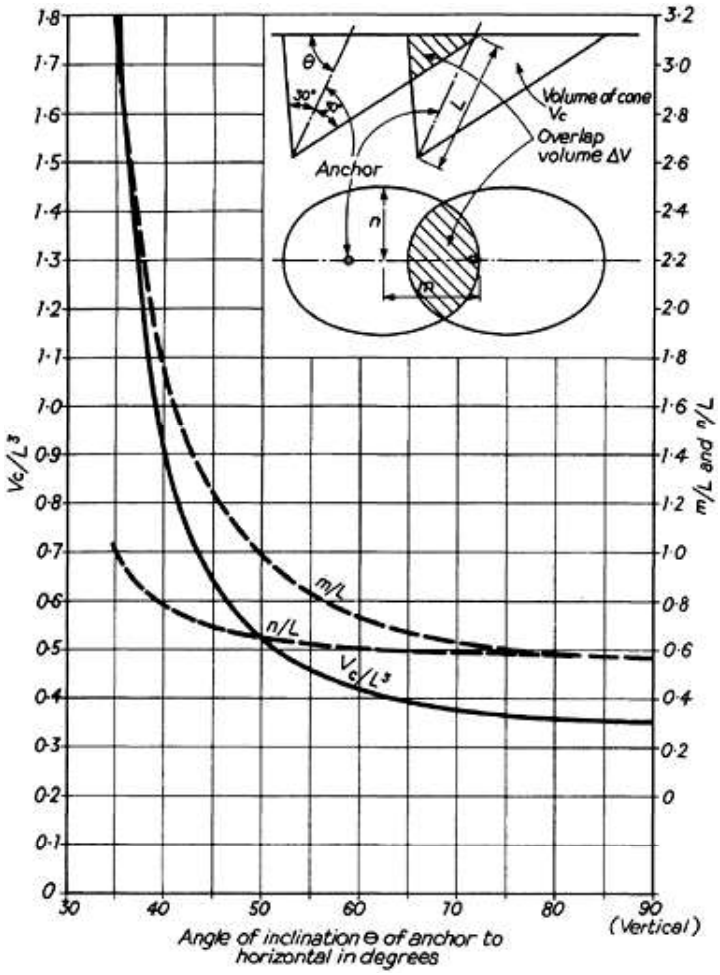


Fig. 6.17 (a) Chart for calculating volumes of single or overlapping cones with vertical or inclined axes

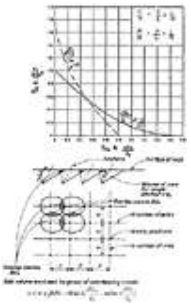


Fig. 6.17 (b) Charts for calculating total volume of rock enclosed by groups of anchors arranged on rectangular grid pattern

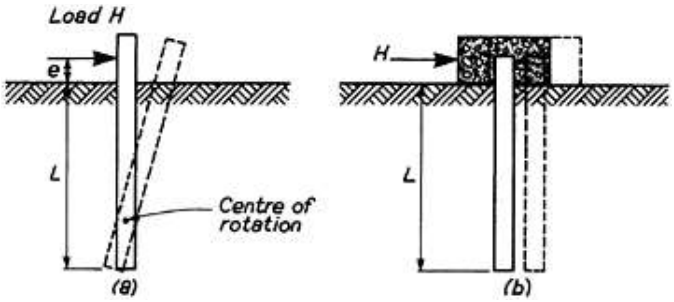


Fig. 6.18 Short vertical pile under horizontal load

the deflection and determines whether the failure mechanism is one of the rotation of a short rigid element, or is due to flexure followed by the failure in bending of a long flexible pile. The type of loading, whether sustained (as in the case of earth pressure transmitted by a retaining wall) or alternating (say, from reciprocating machinery) or pulsating (as from the traffic loading on a bridge pier), influences the degree of yielding of the soil. External influences such as scouring around piles at sea-bed level, or the seasonal shrinkage of clay soils away from the upper part of the pile shaft, affect the resistance of the soil at a shallow depth.

The simpler methods of calculating ultimate resistance and deflection under lateral loads are presented in the following sections of this chapter. No attempt is made to give the complete theoretical basis

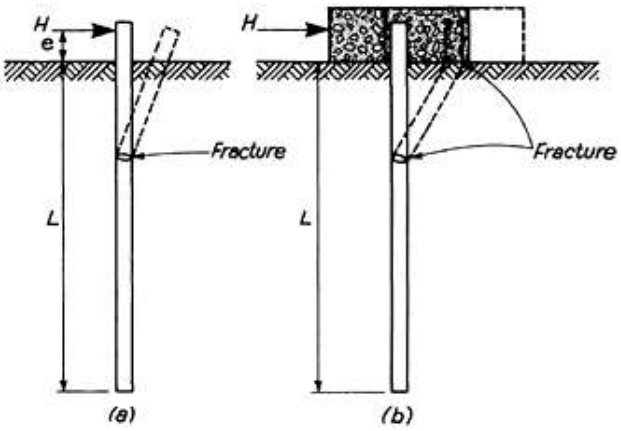


*Fig. 6.19 Long vertical pile under horizontal loa*d

(a) Free head
(b) Fixed head

of these methods, which have been reviewed by Broms[6.13, 6.14] and Elson[6.15]. Various simplifications have been necessary in order to provide simple solutions to complex problems of soil-structure interaction, and the limitations of the methods are stated where these are particularly relevent. Most practical calculations are processes of trial and adjustment, starting with a very simple approach to obtain an approximate measure of the required stiffness and embedment depth of the pile. The process can then be elaborated to some degree to narrow the margin of error, and to provide the essential data for calculating bending moments, shearing forces and deflections at the working load. Very elaborate calculation processes are not justified, because of the non-homogeneity of most natural soil deposits and the disturbance to the soil caused by installing piles. None of these significant factors can be reproduced in the calculation methods.
Eurocode Section 7.8 requires the design of piles to resist transverse loading to be carried out using the same rules as specified for compression loading in Section 7.6 of code (as discussed in Chapter 4 of this book). Where the transverse capacity is calculated from soil test results and pile strength parameters, the capacity is required to be calculated using a compatible set of structural bending moments, shear forces, soil reactions and displacements. The pile strength is required to be determined in accordance with the relevant material Eurocodes for structural design.
For any important foundation structure which has to carry high or sustained lateral loading it is advisable to make field loading tests on trial piles having at least three different shaft lengths, in order to assess the effects of embedment depth and structural stiffness. For less-important structures, or where there is previous experience of pile behaviour to guide the engineer, it may be sufficient to make lateral loading tests on pairs of working piles by jacking or pulling them apart. These tests are rapid and economical to perform (see Section 11.4.4) and provide a reliable check that the design requirements have been met.

6.3.1 Calculating the ultimate resistance to lateral loads

The first step is to determine whether the pile will behave as a short rigid unit or as an infinitely long flexible member. This is done by calculating the stiffness factors R and T for the particular combination of pile and soil. The stiffness factors are governed by the stiffness (EI value) of the pile and the compressibility of the soil. The latter is expressed in terms of a 'soil modulus', which is not constant for any soil type but depends on the width of the pile B and the depth of the particular loaded area of soil being considered. The soil modulus k has been related to Terzaghi's concept of a modulus of horizontal subgrade reaction[6.16]. In the case of a stiff over-consolidated clay, the soil modulus is generally assumed to be constant with depth. For this case

$$\text{stiffness factor } R = \sqrt[4]{\frac{EI}{kB}} \text{ (in units of length)} \qquad \ldots(6.8)$$

Relationships between k and the Terzaghi modulus k1[16.16] and other methods of determining k are given in Section 6.3.3. The modulus k1, is obtained from load-deflection measurements on a 305mm square plate. It is related to the undrained shearing strength of the clay, as shown in Table 6.2.

Table 6.2 Relationship of modulus of subgrade reaction (k1) to untrained shearing strength of stiff overconsolidated clay

Consistency	Firm to stiff	Stiff to very stiff	Hard
Undrained shear strength (c_u) kN/m^2	50–100	100–200	>200
Range of k_1 MN/m^3	15–30	30–60	> 60

For most normally-consolidated clays and for granular soils the soil modulus is assumed to increase linearly with depth, for which

$$\text{stiffness factor } T = \sqrt[5]{\frac{EI}{n_h}} \text{ (in units of length)} \qquad \ldots(6.9)$$

$$\text{where soil modulus } K = n_h \times x/B \qquad \ldots(6.10)$$

Values of the coefficient of modulus variation n_h were obtained directly from lateral loading tests on instrumented piles in submerged sand at Mustang Island, Texas. The tests were made for both static and cyclic loading conditions and the values obtained, as quoted by Reese *et al.*(6.17) were considerably higher than those of Terzaghi(6.16). The investigators recommend that the Mustang Island values should be used for pile design and these are shown together with the Terzaghi values in Figure 6.20.

Other observed values of n_h are as follows:

Soft normally-consolidated clays: 350 to 700kN/m^3.
Soft organic silts: 150kN/m^3.

Having calculated the stiffness factors R or T, the criteria for behaviour as a short rigid pile or as a long elastic pile are related to the embedded length L as follows:

Pile type	Soil modulus	
	Linearly-increasing	Constant
Rigid (free head)	$L \leqslant 2T$	$L \leqslant 2R$
Elastic (free head)	$L \geqslant 4T$	$L \geqslant 3.5R$

Ultimate resistance of short rigid piles

Brinch Hansen's method(6.19) can be used to calculate the ultimate lateral resistance of short rigid piles. The method is a simple one which can be applied both to uniform and layered soils. The resistance

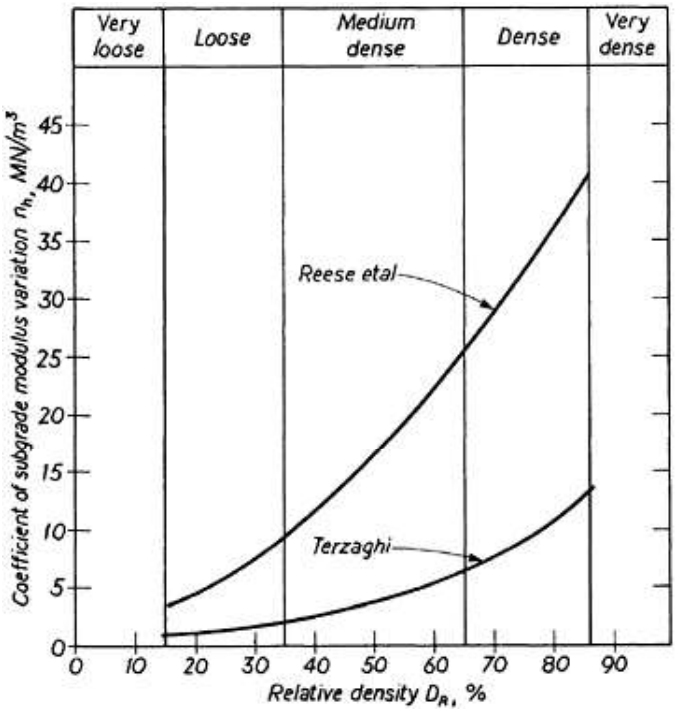


Fig. 6.20 Relationship between coefficient of modulus variation and relative density of sands (after Garassino et al[6-18]*)*

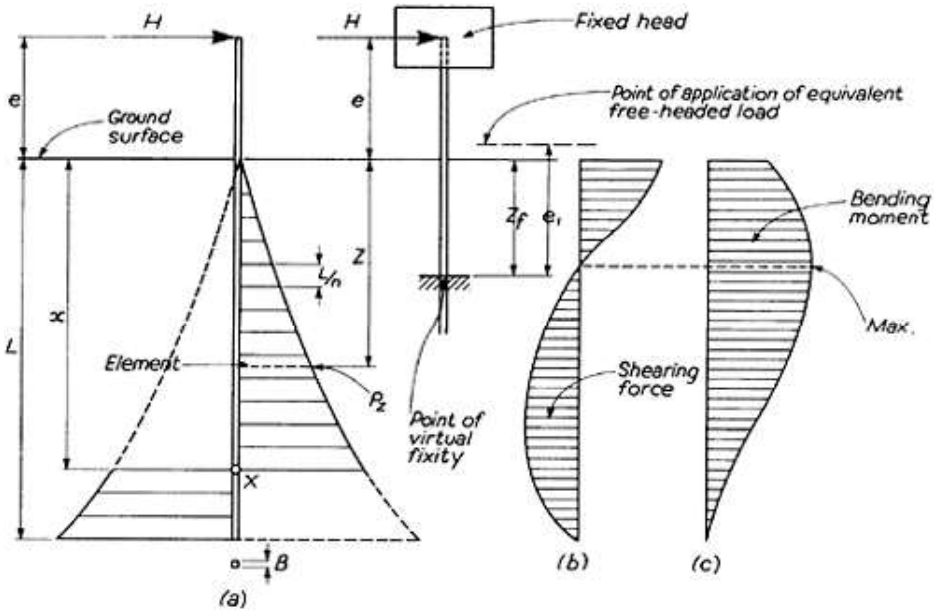


Fig. 6.21 Brinch Hansen's method for calculating ultimate lateral resistance of short piles

(a) Soil reactions
(b) Shearing force diagram
(c) Bending moment diagram

of the rigid unit to rotation about point X in Figure 6.21 a is given by the sum of the moments of the soil resistance above and below this point. The passive resistance diagram is divided into a convenient number n of horizontal elements of depth L/n. The unit passive resistance of an element at a depth z below the ground surface is then given by

$$p_z = p_{oz} K_{qz} + cK_{cz} \qquad \ldots(6.11)$$

where p_{oz} is the effective overburden pressure at depth z, c is the cohesion of the soil at depth z, and K_{qz} and K_{cz} are the passive pressure coefficients for the frictional and cohesive components respectively at depth z. Brinch Hansen[6.19] has established values of K_q and K_c in relation to the depth z and the width of the pile B in the direction of rotation, as shown in Figure 6.22.

The total passive resistance on each horizontal element is $p_z \times \frac{L}{n} \times B$ and, by taking moments about the point of application of the horizontal load,

$$\sum M = \sum_{z=0}^{z=x} p_z \frac{L}{n}(e+z)B - \sum_{z=x}^{z=L} p_z \frac{L}{n}(e+z)B \qquad \ldots(6.12)$$

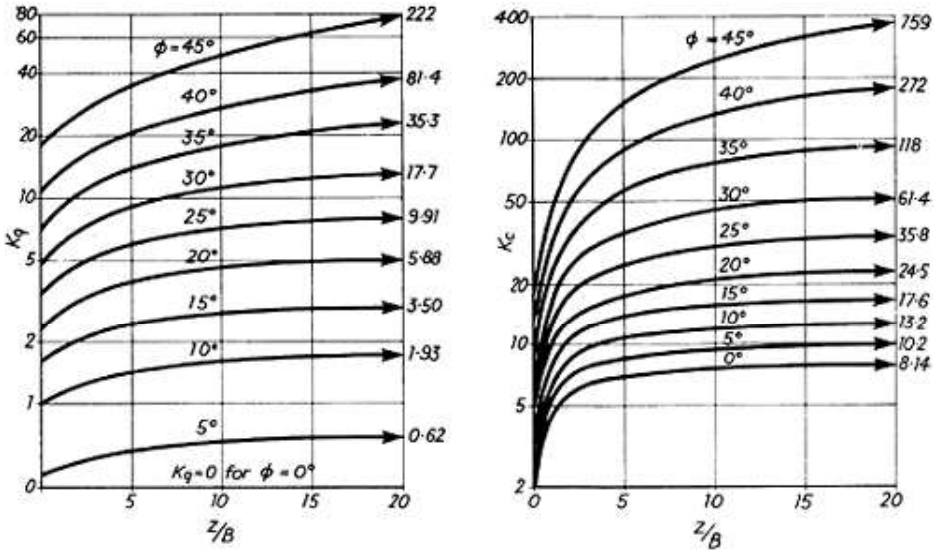


Fig. 6.22 Brinch Hansen's coefficients K_q *and* K_c

The point of rotation at depth x is correctly chosen when $\Sigma M = 0$, *i.e.* when the passive resistance of the soil above the point of rotation balances that below it. Point X is thus determined by a process of trial and adjustment. If the head of the pile carries a moment M instead of a horizontal force, the moment can be replaced by a horizontal force H at a distance e above the ground surface where M is equal *to* $H \times e$.

Where the head of the pile is fixed against rotation, the equivalent height e_1 above ground level of a force H acting on a pile with a free head is given by

$$e_1 = \tfrac{1}{2}(e + z_f) \qquad \ldots(6.13)$$

where e is the height from the ground surface to the point of application of the load at the fixed head of the pile (Figure 6.21a), and z_f is the depth from the ground surface to the point of virtual fixity. The depth z_f is not known at this stage but for practical design purposes it can be taken as 1.5m for a compact granular soil or stiff clay (below the zone of soil shrinkage in the latter case), and 3m for a soft clay or silt. The American Concrete Institute[2.9] recommend that zf should be taken as $1.4R$ for stiff, over-consolidated clays and $1.8T$ for normally consolidated clays, granular soils and silt, and peat (see equations 6.8 and 6.9).

Having obtained the depth to the centre of rotation from equation 6.12, the ultimate lateral resistance of the pile to the horizontal force H_u can be obtained by taking moments about the point of rotation, when

$$H_u(e + x) = \sum_{0}^{x} p_z \frac{L}{n} B(x - z) + \sum_{x}^{x+L} p_z \frac{L}{n} + B(z - x) \qquad \ldots(6.14)$$

The final steps in Brinch Hansen's method are to construct the shearing force and bending moment diagrams (Figure 6.21b and 6.21c). The ultimate bending moment, which occurs at the point of zero shear, should not exceed the ultimate moment of resistance M_u of the pile shaft. The appropriate load factors are applied to the horizontal design force to obtain the ultimate force H_u.

When applying the method to layered soils, assumptions must be made concerning the depth z to obtain K_q and K_c for the soft clay layer, but z is measured from the top of the stiff clay stratum to obtain K_c for this layer, as shown in Figure 6.23.

The undrained shearing strength c_u is used in equation 6.11 for short-term loadings such as wave or ship-berthing forces on a jetty, but the drained effective shearing strength values (c' and) ϕ' are used for long-term sustained loadings such as those on retaining walls. A check should be made to ensure that there is an adequate safety factor for undrained conditions in the early stages of loading. The step-by-step procedure using the Brinch Hansen method is illustrated in worked examples 6.4 and 6.5.

For short-term loading in uniform cohesive $(\phi = 0)$ soils the method of Broms[6.13] is quick and convenient to use. For such soils Broms has assumed that the reaction of the soil on the pile is represented by the simplified diagrams for piles with free and fixed heads shown in Figure 6.24. The zone of zero pressure over the depth $1.5B$ represents the effect of soil shrinkage away from the pile. The ultimate lateral resistance H_u is obtained from Figure 6.25, in which it is related to the undrained shearing strength c_u, the pile width B and the ratio L/B of embedded length to width. From the diagram in Figure 6.24

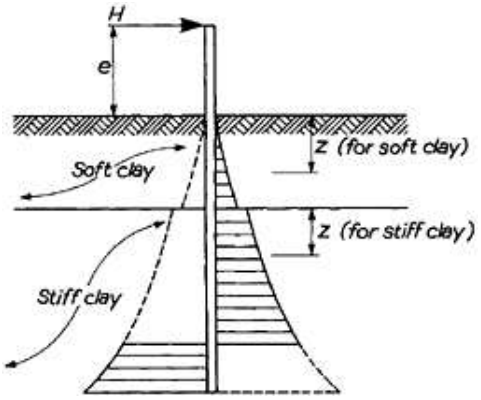


Fig. 6.23 Reactions in layered soil on vertical pile under horizontal load

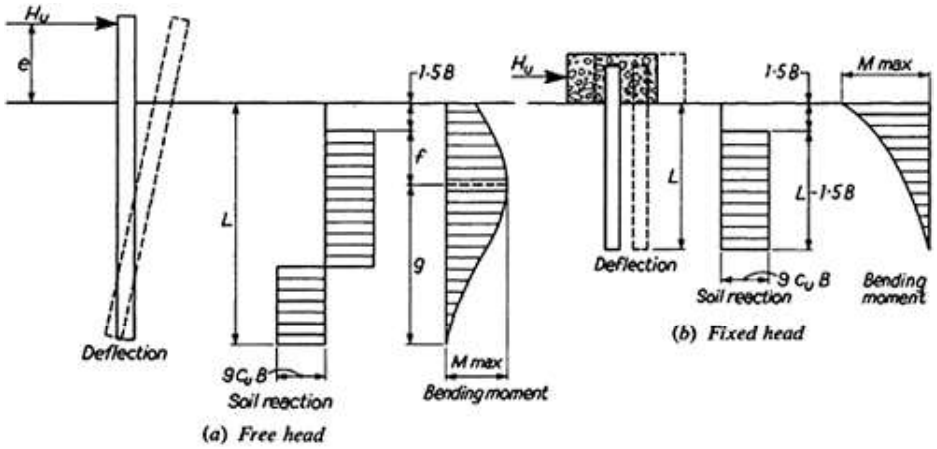


Fig. 6.24 Soil reactions and bending moments for short pile under horizontal load in cohesive soil (after Broms[6.13])

the depth to the point of zero shear *f,* and from this the maximum bending moment for free-headed conditions, can be calculated from

$$f = H/9c_uB \qquad \ldots (6.15)$$

and

$$M_{max} = H(e + 1.5B + 0.5f) \qquad \ldots (6.16)$$

The part of the pile of length *g* resists the bending moment M_{max}, and from equilibrium in short rigid piles,

$$M_{max} = 2.25c_uBg^2 \qquad \ldots (6.17)$$

The short rigid fixed-headed pile (Figure 6.24b) behaves as a simple cantilever carrying a load over part of its length, for which

$$M_{max} = \tfrac{1}{2} \times 9c_uB(L^2 - 2.25B^2) \qquad \ldots (6.18)$$

The ultimate lateral resistance for the fixed-headed pile can also be determined from Figure 6.25.
For cohesionless (c_u=0) soils the distributions of soil reaction and bending moment are as shown in Figure 6.26. At any depth *z* the unit soil reaction on the pile is given by

$$p_z = 3Bp_{oz}K_p \qquad \ldots (6.19)$$

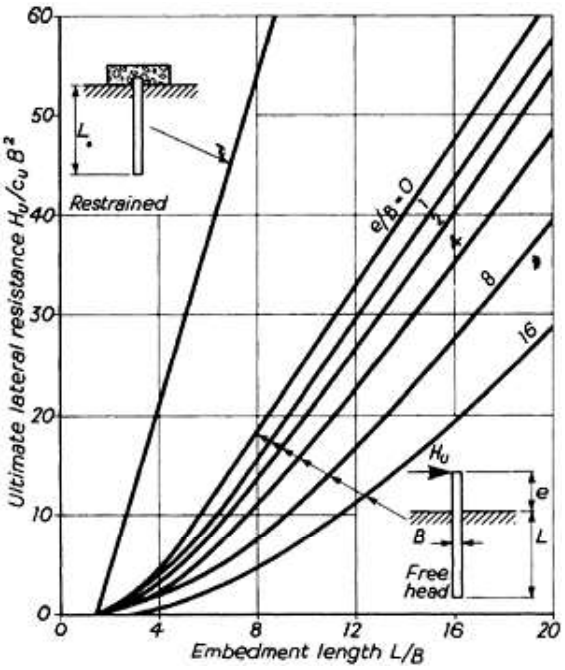


Fig. 6.25 Ultimate lateral resistance of short pile in cohesive soil related to embedded length (after Broms[6.13])

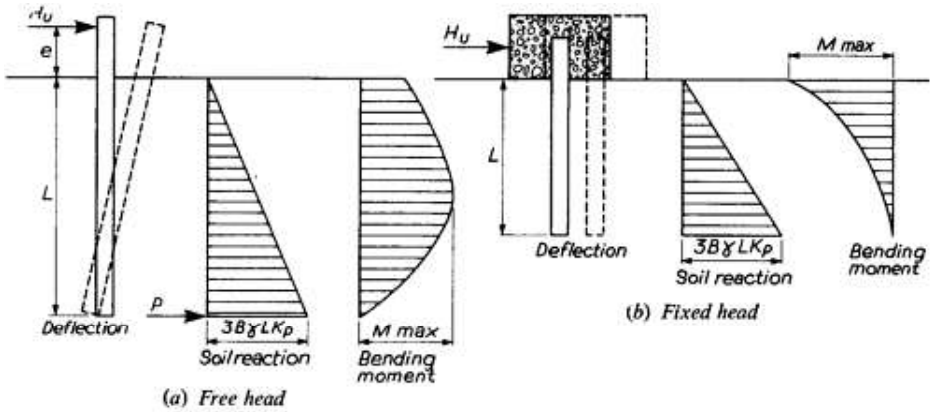


Fig. 6.26 Soil reactions and bending moments for short pile under horizontal load in cohesionless soil (after Brooms[6.14])

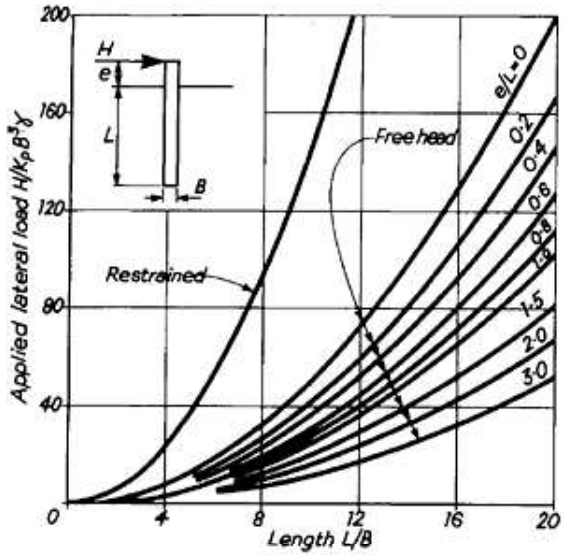


Fig. 6.27 Ultimate lateral resistance of short pile in cohesionless soil related to embedded length (after Broms[6.14])

where B is the width of the pile perpendicular to the direction of rotation, p_{oz} is the effective overburden pressure at a depth z and K_p is Rankine's coefficient of passive pressure, *i.e.*

The high passive resistance at the toe of a pile in a cohesionless soil $K_p = (1 + \sin\phi)/(1 - \sin\phi)$ can be replaced by a single concentrated horizontal force P and, from equation 6.19, the bending moments and shearing forces of the driving and the resistance forces can be equated to obtain the ultimate lateral resistance H_u in a similar manner to that already described for Brinch Hansen's method.

For uniform cohesionless soils, Broms[6.14] has established the graphical relationships for H/kp_B3^{γ} and L/B shown in Figure 6.27, from which the ultimate lateral resistance H_u can be determined. It can also be obtained from

$$H_u = 0.5BL^3K_p\gamma/(e + L) \qquad \ldots(6.20)$$

For short rigid fixed-headed piles in cohesionless soils, failure is due to simple translation (Figure 6.26b), from which

$$H_u = 1.5\mathrm{B}\gamma L^2K_p \qquad \ldots(6.21)$$

Equation 6.21 is valid only when the maximum negative bending moment which develops at the pile head is less than the ultimate resistance moment M_u of the pile at this point. The bending moment is given by

$$M_{max(-ve)} = B\gamma L^3 K_p \qquad \ldots(6.22)$$

Ultimate resistance of long piles

The passive resistance provided by the soil to the yielding of an infinitely long pile is infinite. Thus the ultimate lateral load which can be carried by the pile is determined solely from the ultimate moment of resistance M_u of the pile shaft.

A simple method of calculating the ultimate load, which may be sufficiently accurate for cases of light loading on short or long piles of small to medium width, for which the cross-sectional area is governed by considerations of the relatively-higher compressive loading, is to assume an arbitrary depth *Zf* to the point of virtual fixity. Then from Figure 6.28,

$$\text{ultimate lateral load on free-headed pile } H_u = M_u/(e + z_f) \qquad \ldots(6.23)$$

$$\text{ultimate lateral load on fixed-headed pile } H_u = 2M_u/(e + z_f) \qquad \ldots(6.24)$$

Arbitrary values for z_f which are commonly used are given in the reference to the Brinch Hansen method.

The ultimate moment of resistance of the pile is also taken by Broms[6.13, 6.14] as the criterion of failure of a long pile. A plastic hinge capable of resisting shear is assumed to form at the point of failure of the shaft. The soil reactions and bending moments for cohesive soils are shown in Figure 6.29.

The maximum (positive) bending moment for a free-headed pile is given by

$$M_{max(+ve)} = H(e + 1.5B + 0.5f) \qquad \ldots(6.25)$$

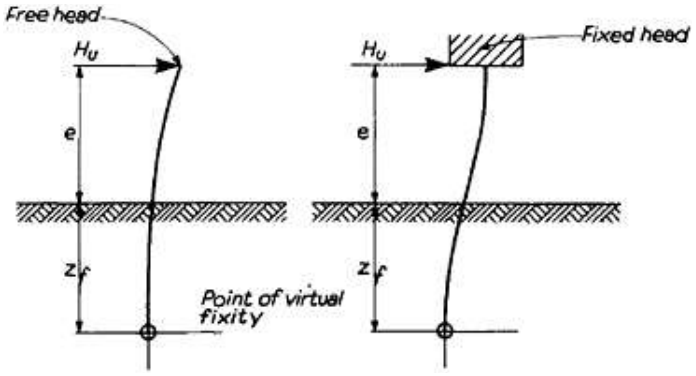


Fig. 6.28 Piles under horizontal load considered as simple cantilevers

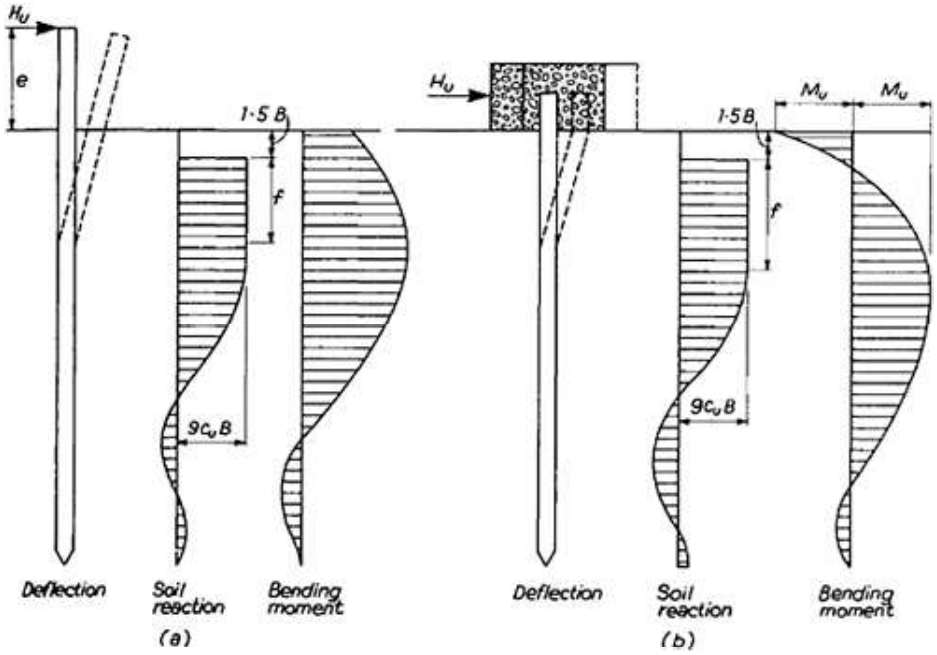


Fig. 6.29 Soil reactions and bending moments for long pile under horizontal load in cohesive soil (after Broms[6.13])

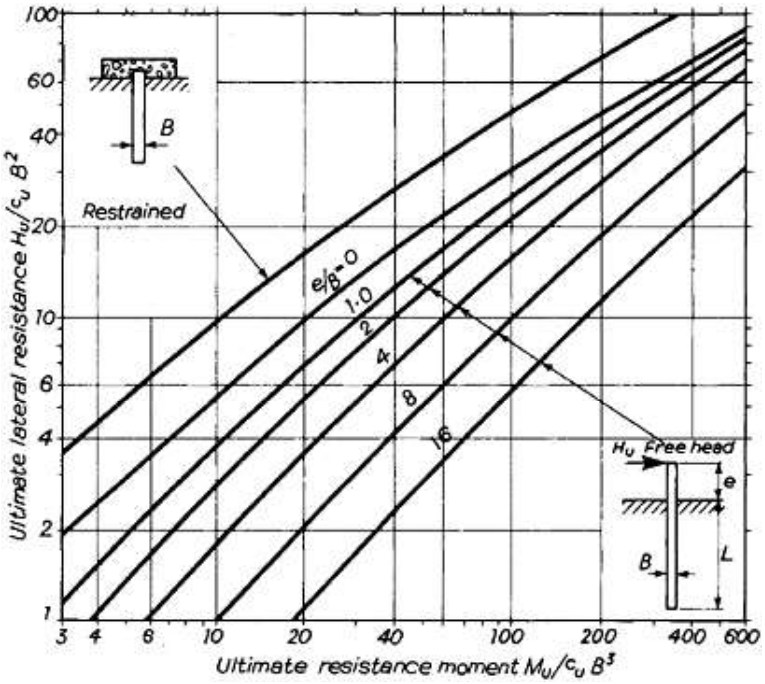


Fig. 6.30 Ultimate lateral resistance of long pile in cohesive soil related to ultimate resistance moment (after Broms[6.13])

where $f=H/9c_uB$. The ultimate value of H is the load at which $M_{max(+ve)}$ is equal to the ultimate resistance moment of the pile shaft. Thus

$$H_u = \frac{M_u}{(e + 1.5B + 0.5f)} \qquad \ldots(6.26)$$

For fixed-headed piles in cohesive soils (Figure 6.29b),

$$H_u = \frac{2M_u}{(1.5B + 0.5f)} \qquad \ldots(6.27)$$

Broms[6.13] has established the graphical relationship for H_u/c_uB^2 and M_u/c_uB^3 shown in Figure 6.30 for free- and fixed-headed piles, from which H_u can be determined.

For long piles in cohesionless soils the soil reactions and bending moments for free-headed piles are shown in Figure 6.31a. The maximum bending moment on the pile shaft occurs at the point where the shearing force is zero. Thus for free-headed piles,

$$f = 0.82\sqrt{\frac{H}{\gamma B K_p}} \qquad \ldots(6.28)$$

The corresponding maximum positive bending moment is given by

$$M_{max(+ve)} = H(e + 0.67f) \qquad \ldots(6.29)$$

For zero bending moment at the pile head, the ultimate lateral load is given by

$$H_u = \frac{M_u}{e + 0.54\sqrt{\frac{H_u}{\gamma B K_p}}} \quad \ldots(6.30)$$

Broms[6.14] has established the graphical relationship between $H/K_p\gamma B^3$ and $M_u/\gamma K_p B^4$ shown in Figure 6.32. These graphs can be used to determine the ultimate lateral load H_u.

The soil reactions and bending moments for fixed-headed piles are shown in Figure 6.31b. The maximum negative bending moment occurs at the pile head and at the ultimate load it is equal to the ultimate moment of resistance of the pile shaft. The ultimate lateral load is given by

$$H_u = \frac{M_{u(+ve)} + M_{u(-ve)}}{e + 0.54\sqrt{\frac{H_u}{\gamma B K_p}}} \quad \ldots(6.31)$$

For a pile of uniform cross-section,

$$H_u = \frac{2M_u}{e + 0.54\sqrt{\frac{H_u}{\gamma B K_p}}} \qquad \ldots(6.32)$$

Broms' graphical relationships for fixed-headed long piles are also shown in Figure 6.32.
It is evident from the worked examples that the ultimate lateral load for a long pile is much higher than that for a short pile of the same cross-section. Therefore when lateral loads are to be carried by vertical piles the aim should be to provide piles of sufficient length to develop their maximum resistance.

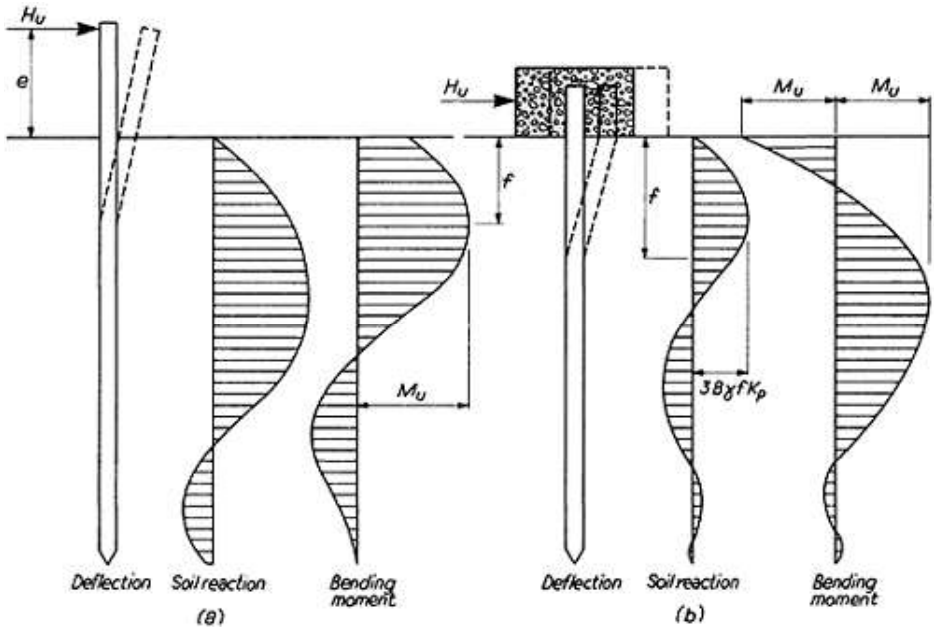


Fig. 6.31 Soil reactions and bending moments for long pile under horizontal load in cohesionless soil (after Broms[6.14])

(a) Free head
(b) Fixed head

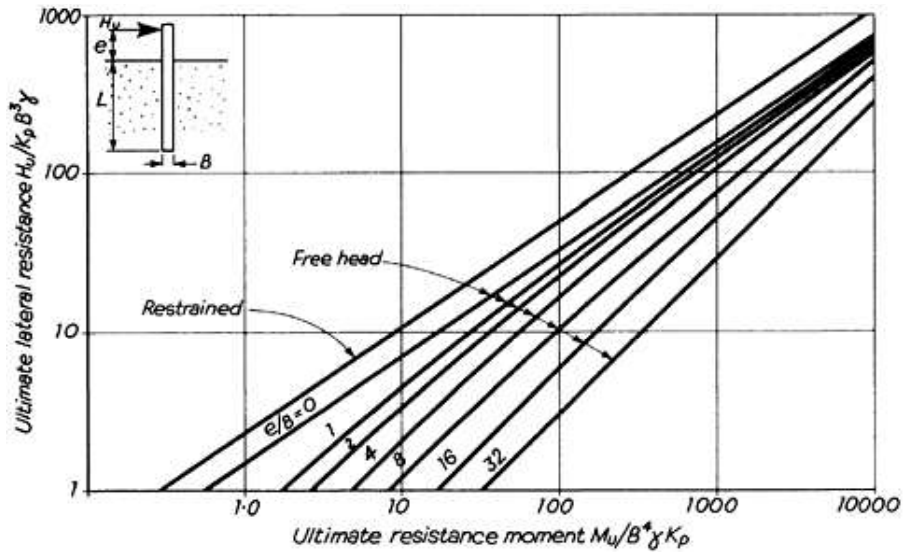


Fig. 6.32 Ultimate lateral resistance of long pile in cohesionless soil related to ultimate resistance moment (after Broms[6.14])

There is, however, no point in taking piles to greater depths than the following.

$L=4T$	(for linearly increasing soil modulus)
$L=3.5R$	(for constant soil modulus and free head)
$L=2R$	(for constant soil modulus and fixed head)

Considerations of the resistance to compressive or uplift loads may require greater penetration depths.
It has already been stated that vertical piles offer poor resistance to lateral loads. However, in some circumstances it may be justifiable to add the resistance provided by the passive resistance of the soil at the end of the pile cap and the friction or cohesion on the embedded sides of the cap. The pile cap resistance can be taken into account when the external loads are transient in character, such as wind gusts and traffic loads, but the resulting elastic deformation of the soil must not be so great as to cause excessive deflection and hence overstressing of the piles. The design of pile caps to resist lateral loading is discussed in 7.9.

6.3.2 Bending and buckling of partly embedded single vertical piles

A partly-embedded vertical pile may be required to carry a vertical load in addition to a lateral load and a bending moment at its head. The stiffness factors R and T as calculated from equations 6.8 and 6.9 have been used by Davisson and Robinson (6.20) to obtain the equivalent length of a free-standing pile with a fixed base, from which the factor of safety against failure due to buckling can be calculated using conventional structural design methods.
A partly-embedded pile carrying a vertical load P, a horizontal load H, and a moment M at a height e above the ground surface is shown in Figure 6.33a. The equivalent height L_e of the fixed-base pile is shown in Figure 6.33b.

For soils having a constant modulus:

$$\text{Depth to point of fixity } z_f = 1.4R \qquad \ldots(6.33)$$

For soils having a linearly-increasing modulus:

$$z_f = 1.8T \qquad \ldots(6.34)$$

The relationships 6.33 and 6.34 are only approximate, but Davisson and Robinson state that they are valid for structural design purposes provided that l_{max}, which is equal to L/R, is greater than 4 for soils having a constant modulus and provided that z_{max}, which is equal to L/T, is greater than 4 for soils having a linearly-increasing modulus. From equations 6.33 and 6.34 the equivalent length L_e of the fixed-base pile (or column) is equal to $e+z_f$ and the critical load for buckling

$$P_{cr} = \frac{\pi^2 EI}{4(e + z_f)^2} \text{ for free-headed conditions} \qquad \ldots(6.35)$$

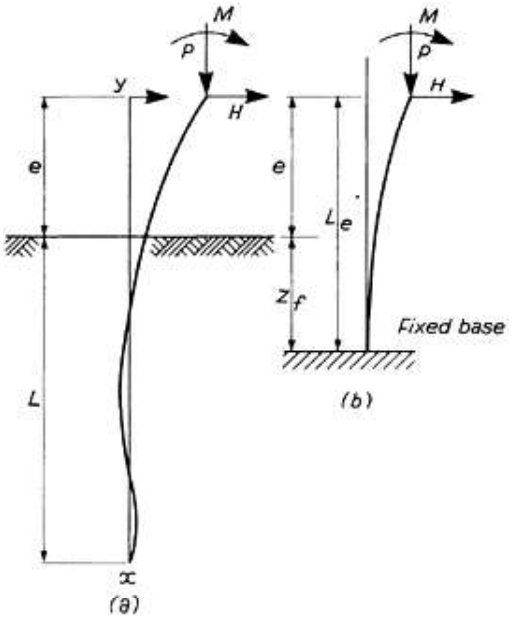


Fig. 6.33 Bending of pile carrying vertical and horizontal loads at head

(a) Partly-embedded pile
(b) Equivalent fixed base pile or column

and

$$P_{cr} = \frac{\pi^2 EI}{(e + z_f)^2} \text{ for fixed- (and translating-) headed conditions} \quad \ldots(6.36)$$

6.3.3 The deflection of vertical piles carrying lateral loads

A simple method which can be used to check that the deflections due to small lateral loads are within tolerable limits and as an approximate check on the more-rigorous methods decribed below, is to assume that the pile is fixed at an arbitrary depth below the ground surface and then to calculate the deflection as for a simple cantilever either free at the head, or fixed at the head but with freedom to translate. Thus from Figure 6.28,

$$\text{deflection at head of free-headed pile } y = \frac{H(e + z_f)^3}{3EI} \quad \ldots(6.37)$$

and

$$\text{deflection at head of fixed-headed pile } y = \frac{H(e + z_f)^3}{12EI} \quad \ldots(6.38)$$

where E is the elastic modulus of the material forming the pile shaft, and I is the moment of inertia of the cross-section of the pile shaft. Depths which may be arbitrarily assumed for z_f are noted in Section 6.3.2.

More-rigorous methods of calculating deflections, which are based on the concept of the modulus of horizontal subgrade reaction, have been described by Broms[6.13, 6.14] as follows.

Piles in cohesive soils
The deflection behaviour depends primarily on the dimensionless length βL where

$$\beta = \sqrt[4]{\frac{kB}{4E_{pile}\, I_{pile}}} \quad \ldots(6.39)$$

A free-headed pile behaves as a rigid (infinitely stiff) element when βL is less than 1.5. For these conditions,

$$\text{lateral deflection at ground surface } y_0 = \frac{4H\left(1 + \frac{1.5e}{L}\right)}{kBL} \quad \ldots(6.40)$$

A fixed-headed pile behaves as a rigid element when βL is less than 0.5, for which

$$y_0 = \frac{H}{kBL} \quad \ldots(6.41)$$

where k is the coefficient of subgrade reaction.
The above equations are not particularly sensitive to the value of k which is, as has already been stated, a variable quantity depending on the depth below the surface. In the case of short rigid piles it is sufficiently accurate to take k as k_1 [the modulus for the 300mm square plate] for the case of a soil with a constant modulus, and as the average value of k_1 over a depth equal to 0.8 βL for a soil with a linearly-increasing modulus. Values of k_1 for cohesive soils are shown in Table 6.2.

For the case of infinitely long flexible piles, for which βL is greater than 2.5, the lateral deflection at the ground surface of the free-headed pile

$$y_0 = \frac{2H\beta(e\beta + 1)}{k_x B} \quad \ldots(6.42)$$

The fixed-headed pile behaves as an infinitely long element when βL is greater than 1.5, and the lateral deflection

$$y_0 = \frac{H\beta}{k_\infty B} \qquad \ldots(6.43)$$

Broms[6.13] has established a graphical relationship between $y_0 kBL/H$ and βL as shown in Figure 6.34, from which the deflection at the ground surface y_0 can be determined.

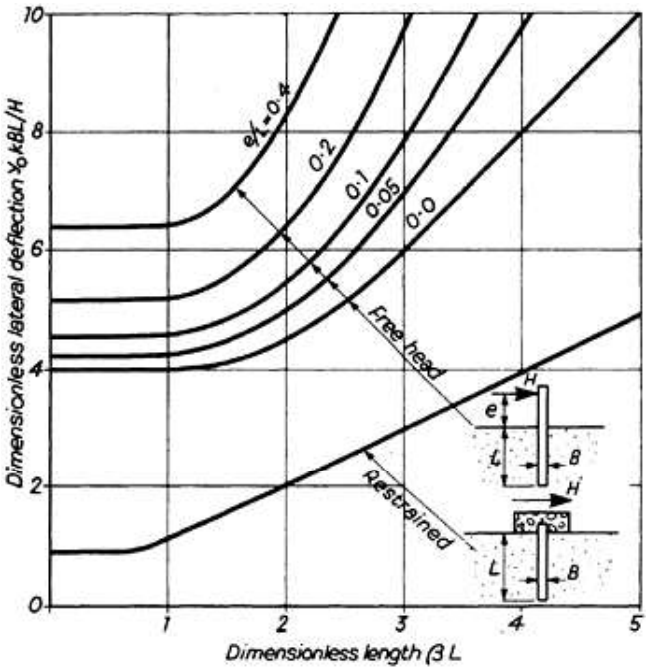


Fig. 6.34 Charts for calculating lateral deflection at ground surface of horizontally loaded pile in cohesive soil (after Broms[6.13])

The coefficient of subgrade reaction k for the infinitely long pile is calculated from

$$k_\infty = \propto K_0/B \qquad \ldots(6.44)$$

where $\propto$ is equal to $0.52\ \sqrt[12]{K_0 B^4/E_{pile}I_{pile}}$ and K_0 is a coefficient of subgrade reaction. Where the Terzaghi modulus is used $K_0 = 0.305\ k_1$.

The calculated deflection is not particularly sensitive to the value of and Broms[6.13] has shown that, for practical purposes, in equation 6.44 can be calculated from

$$\propto = n_1 \times n_2 \qquad \ldots(6.45)$$

where the dimensionless factors n_1 and n_2 are related to the compressive strength of the soil and the material forming the pile respectively. Assuming that the shearing strength of a clay is equal to one-half of the unconfined compressive strength, the values of n_1 are shown in Table 6.3. Values of n_2 for various pile materials are shown in Table 6.4.

Table 6.3 Values of coefficient n_1 (after Broms[6.13])

Shearing strength kN/m^2	Coefficient n_1
<27	0.32
27–107	0.36
>107	0.40

Table 6.4 Values of coefficient n_2 (after Broms[6.13])

Material forming pile	Coefficient n_2
Steel	1.00

Concrete	1.15
Wood	1.30

Where the results of plate loading tests (made with plates set in a vertical position) are available, the coefficient K_0 in equation 6.44 can be obtained from the secant modulus of the soil. A secant modulus E_{50}, corresponding to a pressure on the soil of 50% of the pressure at the failure of the plate, is used in equation

$$K_0 = 1.67E_{50} \quad \ldots(6.46)$$

The above equations for calculating deflections are all based on elastic theory and take no account of deflections caused by the long-term consolidation of the soil under sustained lateral loading. This can be done by using methods based on elastic continuum theory described in Section 6.3.8. Methods of calculating pile displacements in conditions of a yielding soil are described in Section 6.3.5.

Piles in cohesionless soils

Broms[6.14] has related the deflection of a pile embedded in a cohesionless soil carrying a horizontal load H to the dimensionless depth of embedment ηL where

$$\eta = \sqrt[5]{n_h/EI} \qquad \ldots(6.47)$$

Values of the relationship between $y_0(EI)^{3/5}n_h^{2/5}/HL$ and ηL for various values of e/L are shown in Figure 6.35. These curves show that for values of ηL of less than 2 the pile behaves as an infinitely stiff member, while when ηL is greater than 4 the pile behaves as an infinitely long member for which the deflections at the ground surface are unaffected by the depth of embedment.

The deflections y_0 of the pile at ground level can be calculated directly without recourse to Figure 6.35 from the following equations.

Short pile *($\eta L < 2$)* with free head:

$$y_0 = \frac{18H\left(1 + \dfrac{1.33e}{L}\right)}{L^2 n_h} \qquad \ldots(6.48)$$

Short pile $(\eta L < 2)$ with fixed head:

$$y_0 = \frac{2H}{L^2 n_h} \qquad \ldots(6.49)$$

Long pile $(\eta L > 4)$ with free head:

$$y_0 = \frac{2.4H(1 + 0.67e\eta)}{n_h^{3/5}(EI)^{2/5}} \qquad \ldots(6.50)$$

Long pile $(\eta L > 4)$ with fixed head:

$$y_0 = \frac{0.93H}{n_h^{3/5}(EI)^{2/5}} \qquad \ldots(6.51)$$

Values of the coefficient of modulus variation n_h are obtained from Figure 6.20. Broms states that equations 6.49 and 6.50 are not applicable to very short piles, *i.e.* with a length of less than 4 diameters, because of the restraining effect of soil friction beneath the toes of such piles.

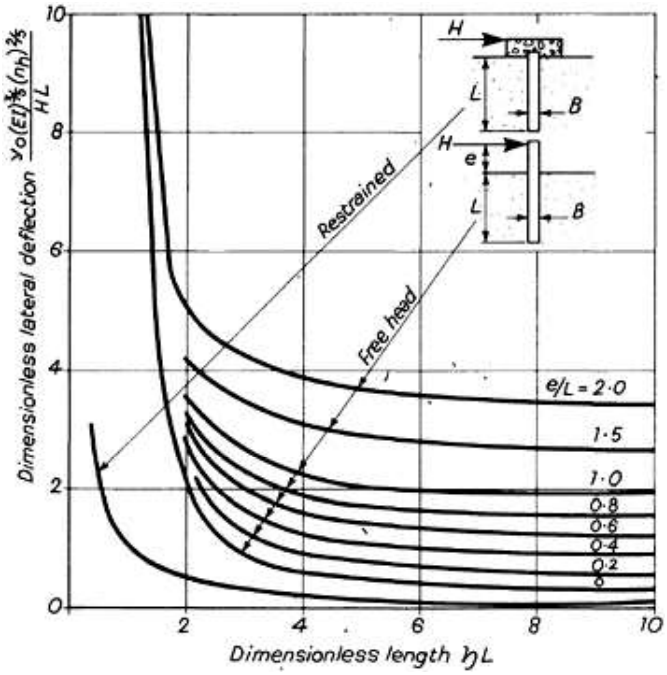


Fig. 6.35 Charts for calculating lateral deflection at ground surface of horizontally loaded pile in cohesionless soil (after Broms[6.14])